Data Analytics (DA)

Module -2

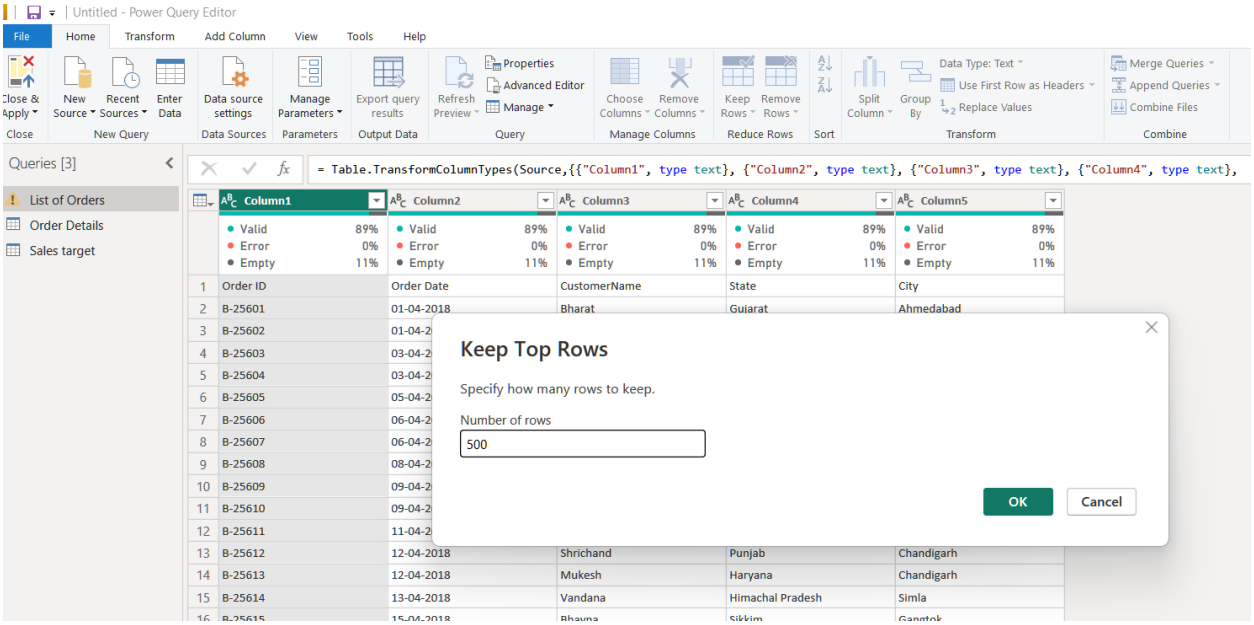
Power BI Assignment 1 – Data Transformation & Data Modeling

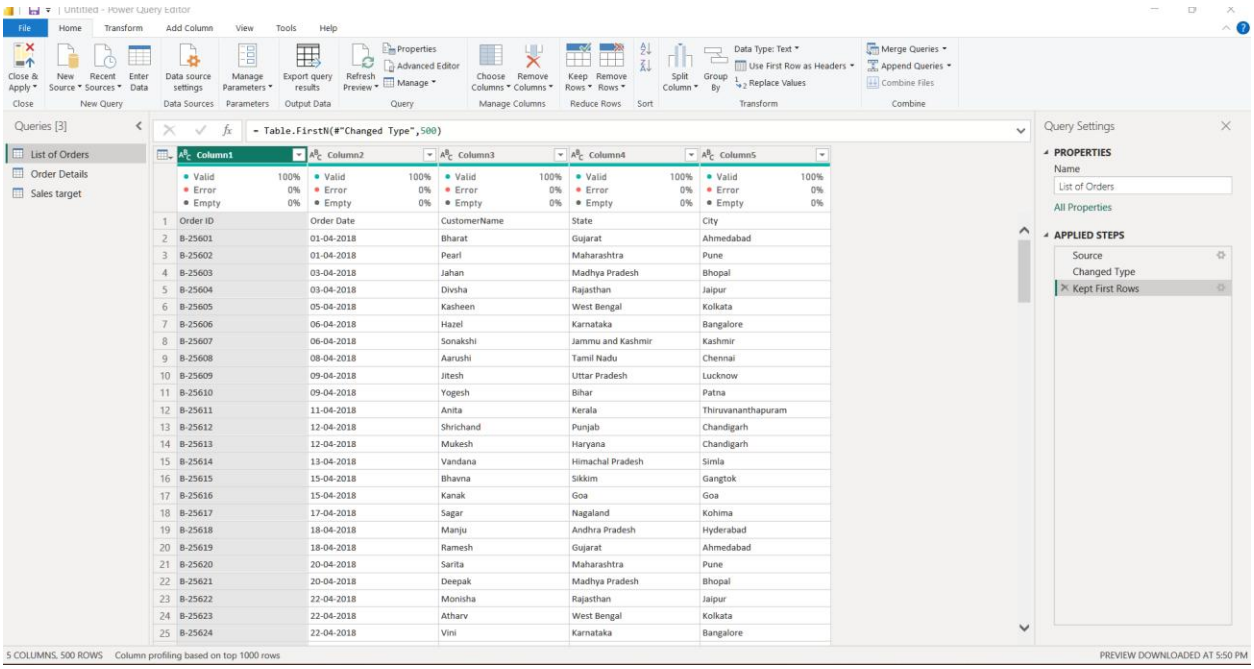
Data Import & Transformation:

Imported List of Orders.csv, Order Details.csv, and Sales Target.csv into Power BI using the Get Data option. Opened all datasets in Power Query Editor through the Transform Data feature for further data cleaning and transformation.

Row Limiting & Data Types:

Restricted the List of Orders dataset to the first 500 rows using the Keep Top Rows option in Power Query Editor. Converted the Order Date column to Date data type and changed the Amount and Target columns to Fixed Decimal Number format for accurate analysis and calculations.




Untitled - Power Query Editor
File Home Transform Add Column View Tools Help
Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files
Close New Query Data Sources Parameters Output Data Query Manage Columns Reduce Rows Sort Transform Combine
Queries [3]
List of Orders
Order Details
Sales target
= Table.FirstN(#"Changed Type",500)
Column1 Column2 Column3 Column4 Column5
Valid 100% Error 0% Empty 0%
Order ID Order Date CustomerName State City
B-25601 01-04-2018 Bharat Gujarat Ahmedabad
B-25602 01-04-2018 Pearl Maharashtra Pune
B-25603 03-04-2018 Jahan Madhya Pradesh Bhopal
B-25604 03-04-2018 Divsha Rajasthan Jaipur
B-25605 05-04-2018 Kasheen West Bengal Kolkata
B-25606 06-04-2018 Hazel Karnataka Bangalore
B-25607 06-04-2018 Sonakshi Jammu and Kashmir Kashmir
B-25608 08-04-2018 Aarushi Tamil Nadu Chennai
B-25609 09-04-2018 Jitesh Uttar Pradesh Lucknow
B-25610 09-04-2018 Yogesh Bihar Patna
B-25611 11-04-2018 Anita Kerala Thiruvananthapuram
B-25612 12-04-2018 Shrichand Punjab Chandigarh
B-25613 12-04-2018 Mukesh Haryana Chandigarh
B-25614 13-04-2018 Vandana Himachal Pradesh Simla
B-25615 15-04-2018 Bhavna Sikkim Gangtok
B-25616 15-04-2018 Kanak Goa Goa
B-25617 17-04-2018 Sagar Nagaland Kohima
B-25618 18-04-2018 Manju Andhra Pradesh Hyderabad
B-25619 18-04-2018 Ramesh Gujarat Ahmedabad
B-25620 20-04-2018 Sarita Maharashtra Pune
B-25621 20-04-2018 Deepak Madhya Pradesh Bhopal
B-25622 22-04-2018 Monisha Rajasthan Jaipur
B-25623 22-04-2018 Atharv West Bengal Kolkata
B-25624 22-04-2018 Vini Karnataka Bangalore
Query Settings
PROPERTIES
Name
List of Orders
All Properties
APPLIED STEPS
Source
Changed Type
Kept First Rows
5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows
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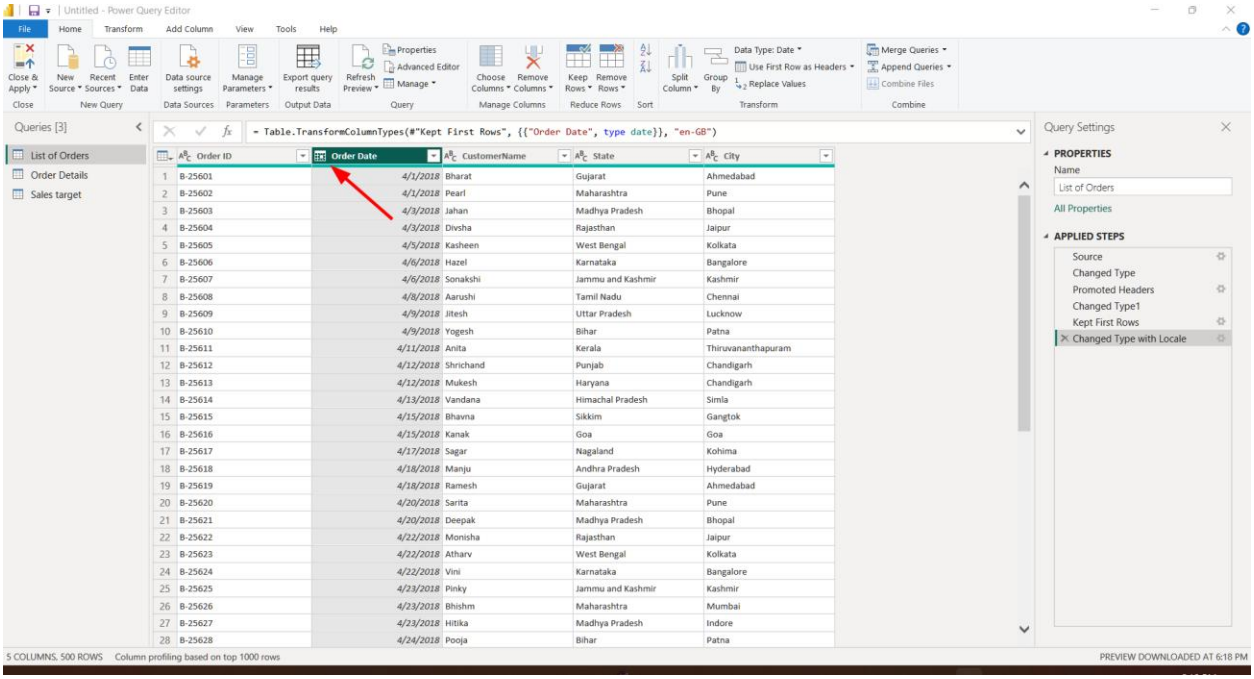
Untitled - Power Query Editor
File Home Transform Add Column View Tools Help
Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Date Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files
Close New Query Data Sources Parameters Output Data Query Manage Columns Reduce Rows Sort Transform Combine
Queries [3]
List of Orders
Order Details
Sales target
= Table.TransformColumnTypes(#"Kept First Rows", {{"Order Date", type date}}, "en-GB")
Order ID Order Date CustomerName State City
B-25601 4/1/2018 Bharat Gujarat Ahmedabad
B-25602 4/1/2018 Pearl Maharashtra Pune
B-25603 4/3/2018 Jahan Madhya Pradesh Bhopal
B-25604 4/3/2018 Divsha Rajasthan Jaipur
B-25605 4/5/2018 Kasheen West Bengal Kolkata
B-25606 4/6/2018 Hazel Karnataka Bangalore
B-25607 4/6/2018 Sonakshi Jammu and Kashmir Kashmir
B-25608 4/8/2018 Aarushi Tamil Nadu Chennai
B-25609 4/9/2018 Jitesh Uttar Pradesh Lucknow
B-25610 4/9/2018 Yogesh Bihar Patna
B-25611 4/11/2018 Anita Kerala Thiruvananthapuram
B-25612 4/12/2018 Shrichand Punjab Chandigarh
B-25613 4/12/2018 Mukesh Haryana Chandigarh
B-25614 4/13/2018 Vandana Himachal Pradesh Simla
B-25615 4/15/2018 Bhavna Sikkim Gangtok
B-25616 4/15/2018 Kanak Goa Goa
B-25617 4/17/2018 Sagar Nagaland Kohima
B-25618 4/18/2018 Manju Andhra Pradesh Hyderabad
B-25619 4/18/2018 Ramesh Gujarat Ahmedabad
B-25620 4/20/2018 Sarita Maharashtra Pune
B-25621 4/20/2018 Deepak Madhya Pradesh Bhopal
B-25622 4/22/2018 Monisha Rajasthan Jaipur
B-25623 4/22/2018 Atharv West Bengal Kolkata
B-25624 4/22/2018 Vini Karnataka Bangalore
B-25625 4/23/2018 Pinky Jammu and Kashmir Kashmir
B-25626 4/23/2018 Bhishm Maharashtra Mumbai
B-25627 4/23/2018 Hitika Madhya Pradesh Indore
B-25628 4/24/2018 Pooja Bihar Patna
Query Settings
PROPERTIES
Name
List of Orders
All Properties
APPLIED STEPS
Source
Changed Type
Promoted Headers
Changed Type1
Kept First Rows
Changed Type with Locale
5 COLUMNS, 500 ROWS Column profiling based on top 1000 rows
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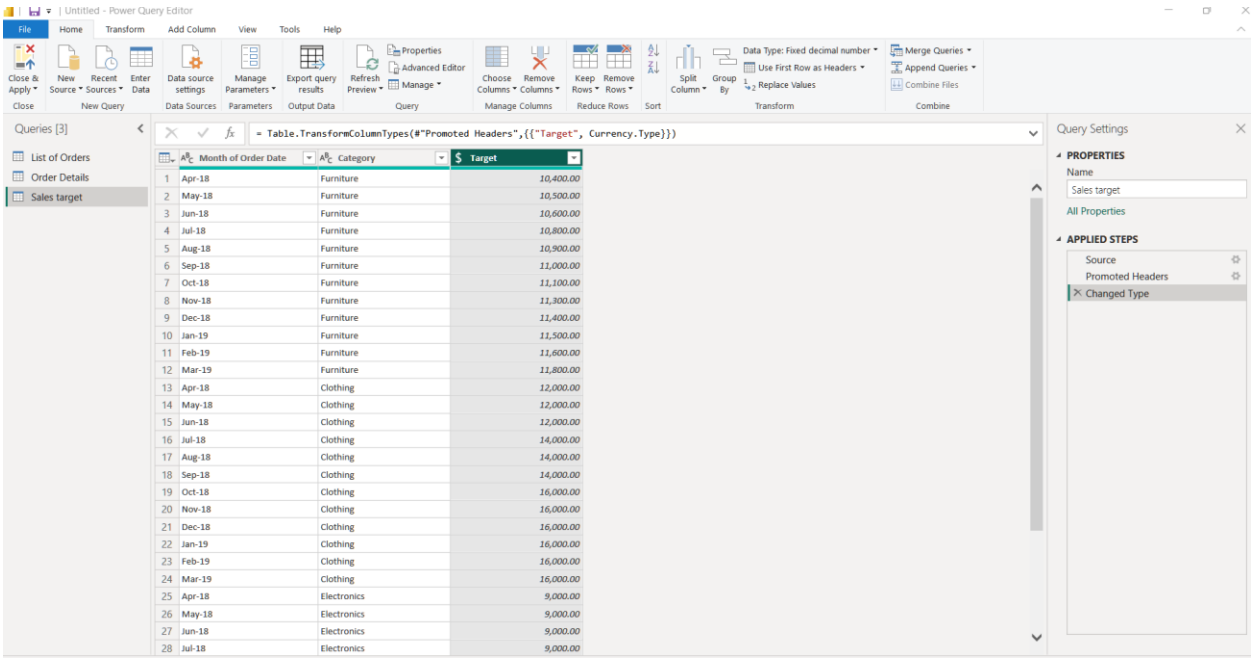
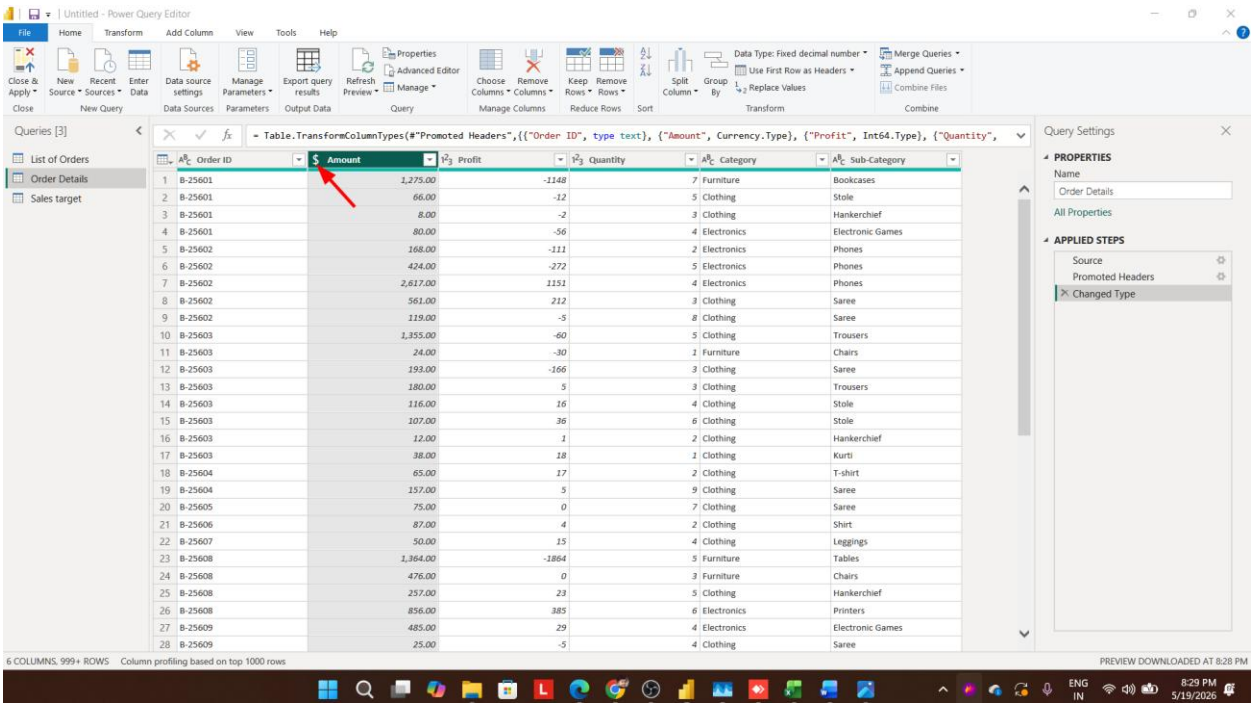


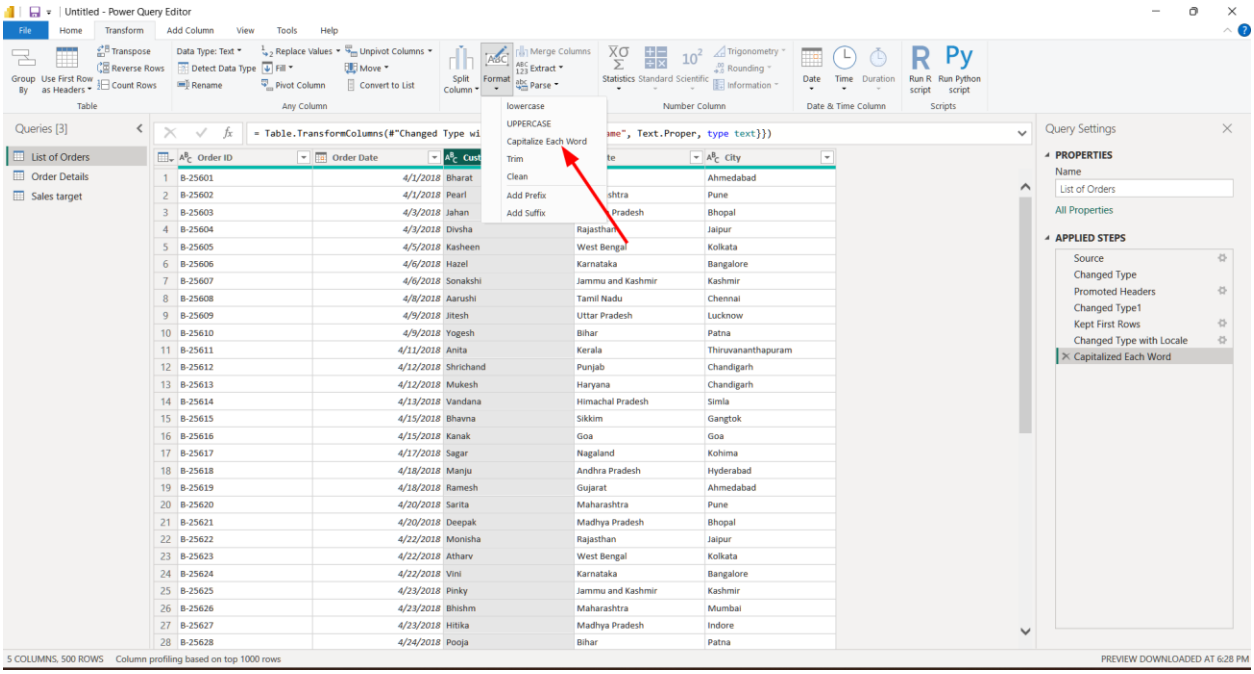
	Month of Order Date	Category	Target
1	Apr-18	Furniture	10,400.00
2	May-18	Furniture	10,500.00
3	Jun-18	Furniture	10,600.00
4	Jul-18	Furniture	10,800.00
5	Aug-18	Furniture	10,900.00
6	Sep-18	Furniture	11,000.00
7	Oct-18	Furniture	11,100.00
8	Nov-18	Furniture	11,300.00
9	Dec-18	Furniture	11,400.00
10	Jan-19	Furniture	11,500.00
11	Feb-19	Furniture	11,600.00
12	Mar-19	Furniture	11,800.00
13	Apr-18	Clothing	12,000.00
14	May-18	Clothing	12,000.00
15	Jun-18	Clothing	12,000.00
16	Jul-18	Clothing	14,000.00
17	Aug-18	Clothing	14,000.00
18	Sep-18	Clothing	14,000.00
19	Oct-18	Clothing	16,000.00
20	Nov-18	Clothing	16,000.00
21	Dec-18	Clothing	16,000.00
22	Jan-19	Clothing	16,000.00
23	Feb-19	Clothing	16,000.00
24	Mar-19	Clothing	16,000.00
25	Apr-18	Electronics	9,000.00
26	May-18	Electronics	9,000.00
27	Jun-18	Electronics	9,000.00
28	Jul-18	Electronics	9,000.00



	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601	1,275.00	-1148	7	Furniture	Bookcases
2	B-25601	66.00	-12	5	Clothing	Stole
3	B-25601	8.00	-2	3	Clothing	Hankerchief
4	B-25601	80.00	-56	4	Electronics	Electronic Games
5	B-25602	168.00	-111	2	Electronics	Phones
6	B-25602	424.00	-272	5	Electronics	Phones
7	B-25602	2,617.00	1151	4	Electronics	Phones
8	B-25602	561.00	212	3	Clothing	Saree
9	B-25602	119.00	-5	8	Clothing	Saree
10	B-25603	1,355.00	-60	5	Clothing	Trousers
11	B-25603	24.00	-30	1	Furniture	Chairs
12	B-25603	193.00	-166	3	Clothing	Saree
13	B-25603	180.00	5	3	Clothing	Trousers
14	B-25603	116.00	16	4	Clothing	Stole
15	B-25603	107.00	36	6	Clothing	Stole
16	B-25603	12.00	1	2	Clothing	Hankerchief
17	B-25603	38.00	18	1	Clothing	Kurti
18	B-25604	65.00	17	2	Clothing	T-shirt
19	B-25604	157.00	5	9	Clothing	Saree
20	B-25605	75.00	0	7	Clothing	Saree
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1864	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25	B-25608	257.00	23	5	Clothing	Hankerchief
26	B-25608	856.00	385	6	Electronics	Printers
27	B-25609	485.00	29	4	Electronics	Electronic Games
28	B-25609	25.00	-5	4	Clothing	Saree

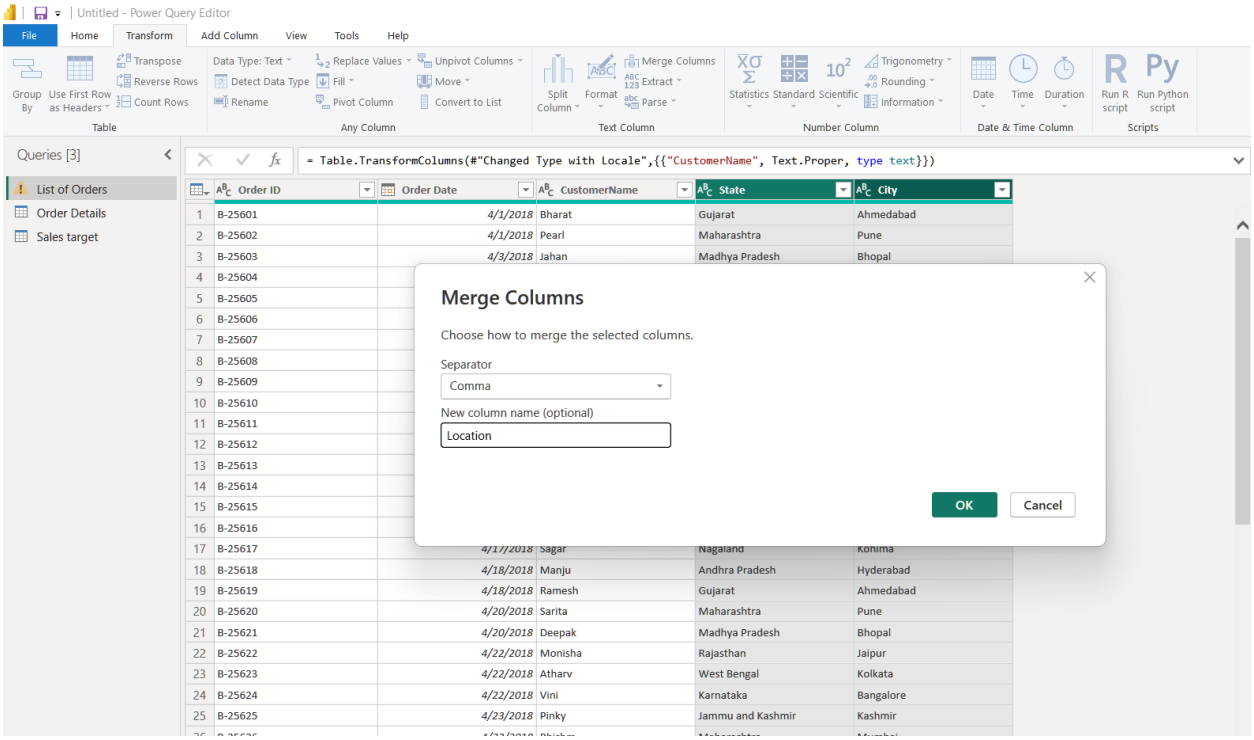
Text Formatting:

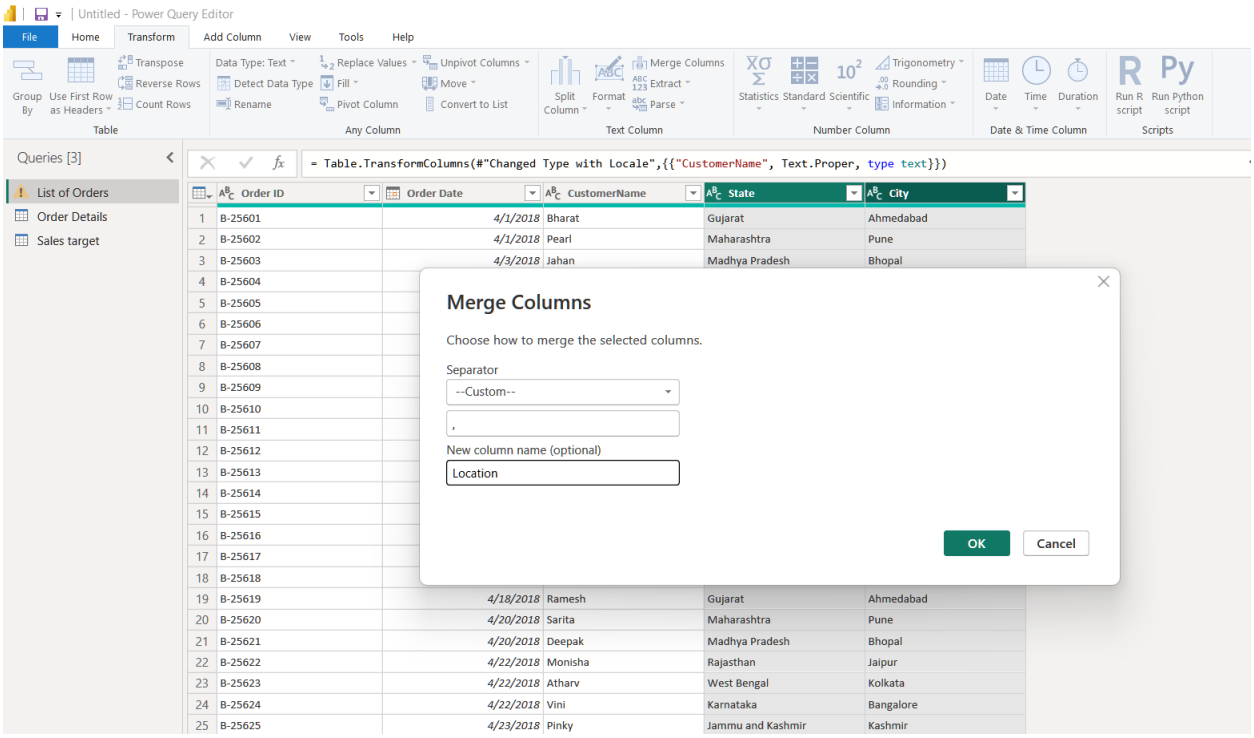
Formatted the Customer Name column using the Capitalize Each Word option in Power Query Editor to maintain consistent capitalization and improve data quality.

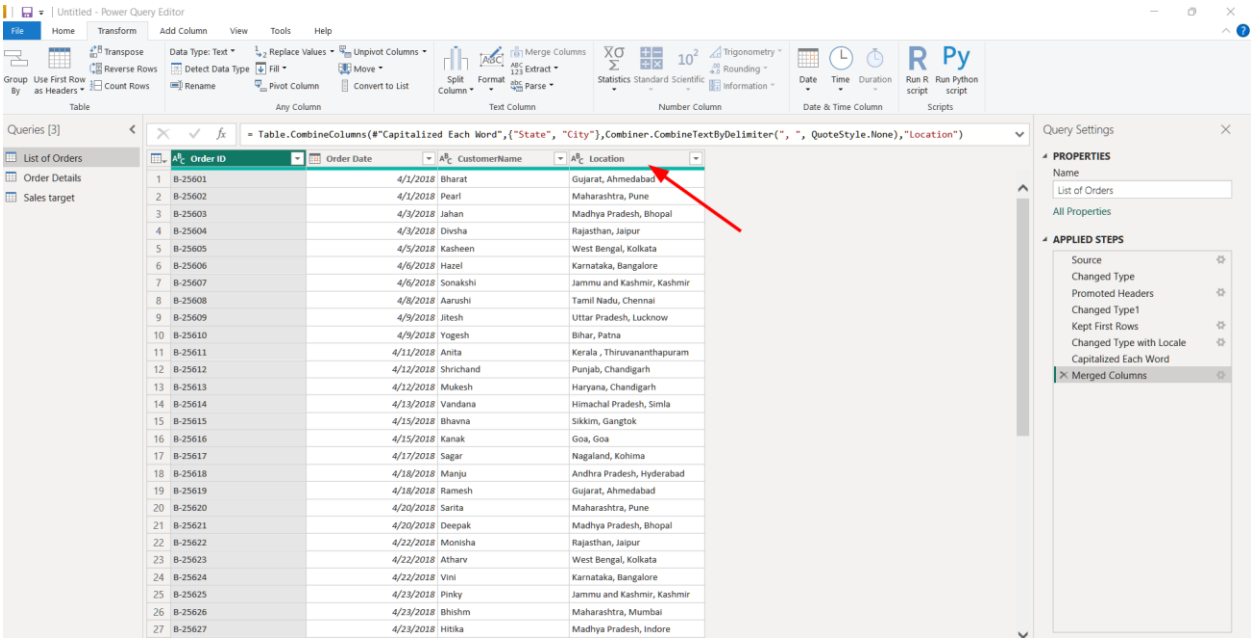


Column Creation:

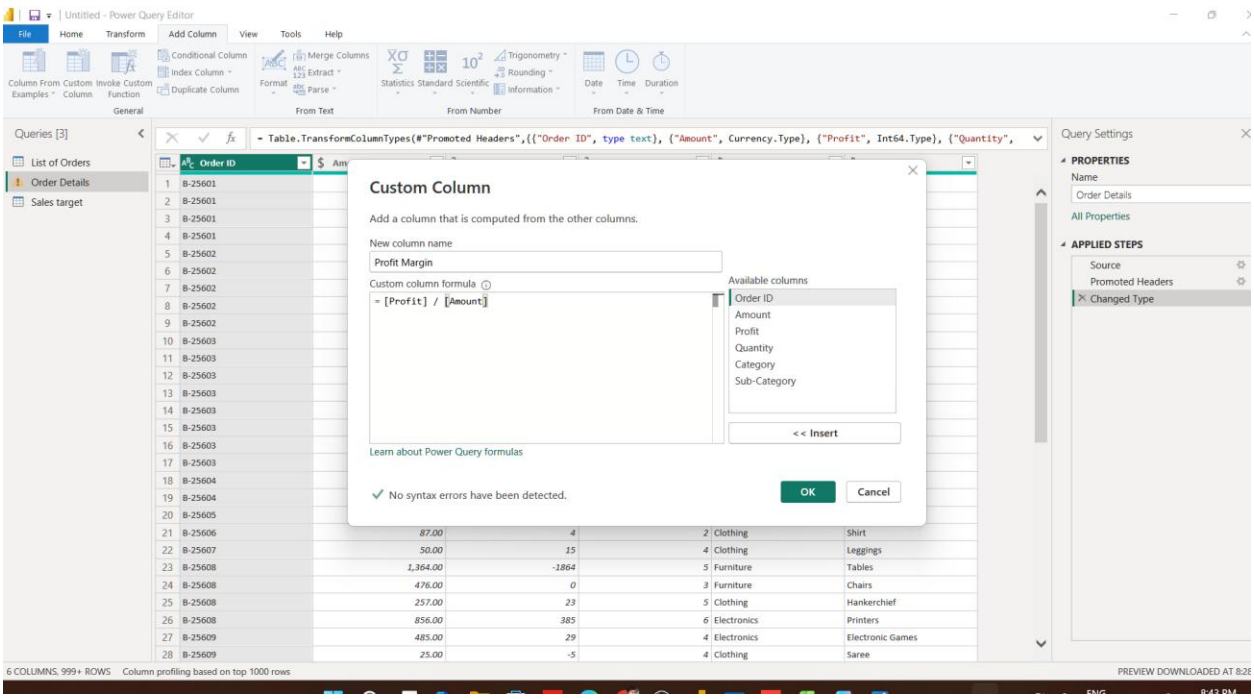
Merged the City and State columns to create a Location field in the format “City, State”. Added a custom column named Profit Margin using the formula Profit divided by Amount and formatted the result as a percentage for profitability analysis.



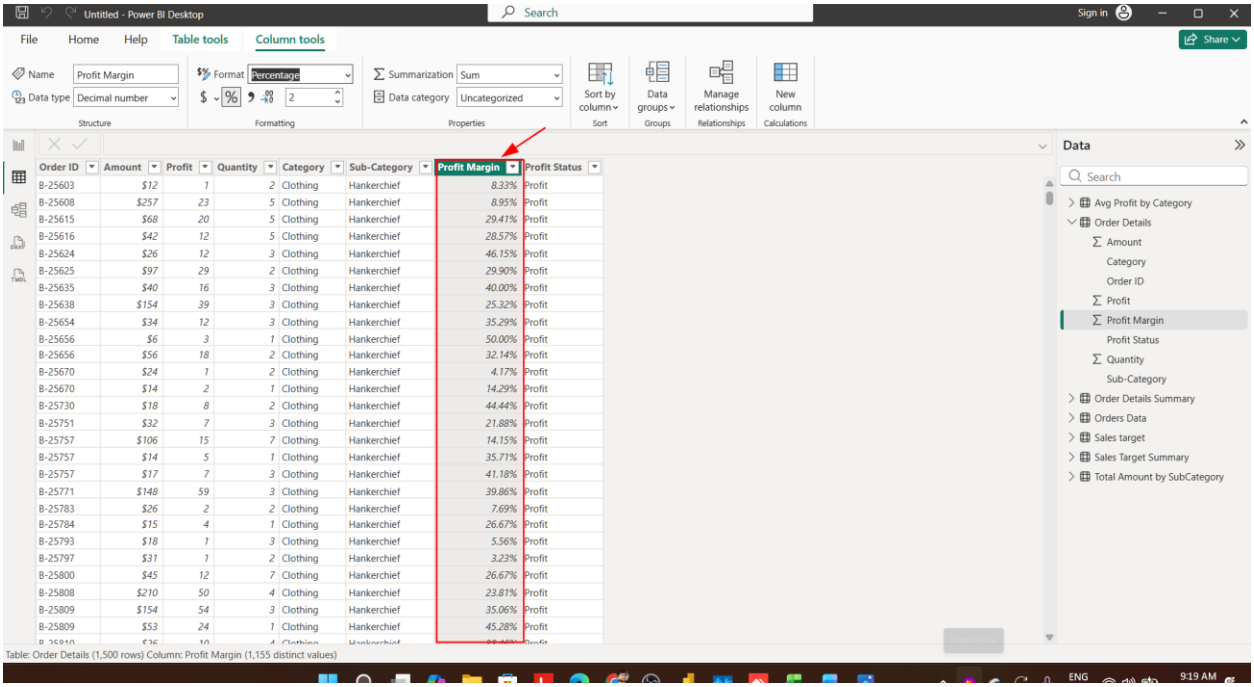




	Order ID	Order Date	CustomerName	Location
1	B-25601	4/1/2018	Bharat	Gujarat, Ahmedabad
2	B-25602	4/1/2018	Pearl	Maharashtra, Pune
3	B-25603	4/3/2018	Jahan	Madhya Pradesh, Bhopal
4	B-25604	4/3/2018	Divsha	Rajasthan, Jaipur
5	B-25605	4/5/2018	Kasheen	West Bengal, Kolkata
6	B-25606	4/6/2018	Hazel	Karnataka, Bangalore
7	B-25607	4/6/2018	Sonakshi	Jammu and Kashmir, Kashmir
8	B-25608	4/8/2018	Aarushi	Tamil Nadu, Chennai
9	B-25609	4/9/2018	Jitesh	Uttar Pradesh, Lucknow
10	B-25610	4/9/2018	Yogesh	Bihar, Patna
11	B-25611	4/11/2018	Anita	Kerala , Thiruvananthapuram
12	B-25612	4/12/2018	Shrichand	Punjab, Chandigarh
13	B-25613	4/12/2018	Mukesh	Haryana, Chandigarh
14	B-25614	4/13/2018	Vandana	Himachal Pradesh, Simla
15	B-25615	4/15/2018	Bhavna	Sikkim, Gangtok
16	B-25616	4/15/2018	Kanak	Goa, Goa
17	B-25617	4/17/2018	Sagar	Nagaland, Kohima
18	B-25618	4/18/2018	Manju	Andhra Pradesh, Hyderabad
19	B-25619	4/18/2018	Ramesh	Gujarat, Ahmedabad
20	B-25620	4/20/2018	Sarita	Maharashtra, Pune
21	B-25621	4/20/2018	Deepak	Madhya Pradesh, Bhopal
22	B-25622	4/22/2018	Monisha	Rajasthan, Jaipur
23	B-25623	4/22/2018	Atharv	West Bengal, Kolkata
24	B-25624	4/22/2018	Vini	Karnataka, Bangalore
25	B-25625	4/23/2018	Pinky	Jammu and Kashmir, Kashmir
26	B-25626	4/23/2018	Bhishm	Maharashtra, Mumbai
27	B-25627	4/23/2018	Hitika	Madhya Pradesh, Indore

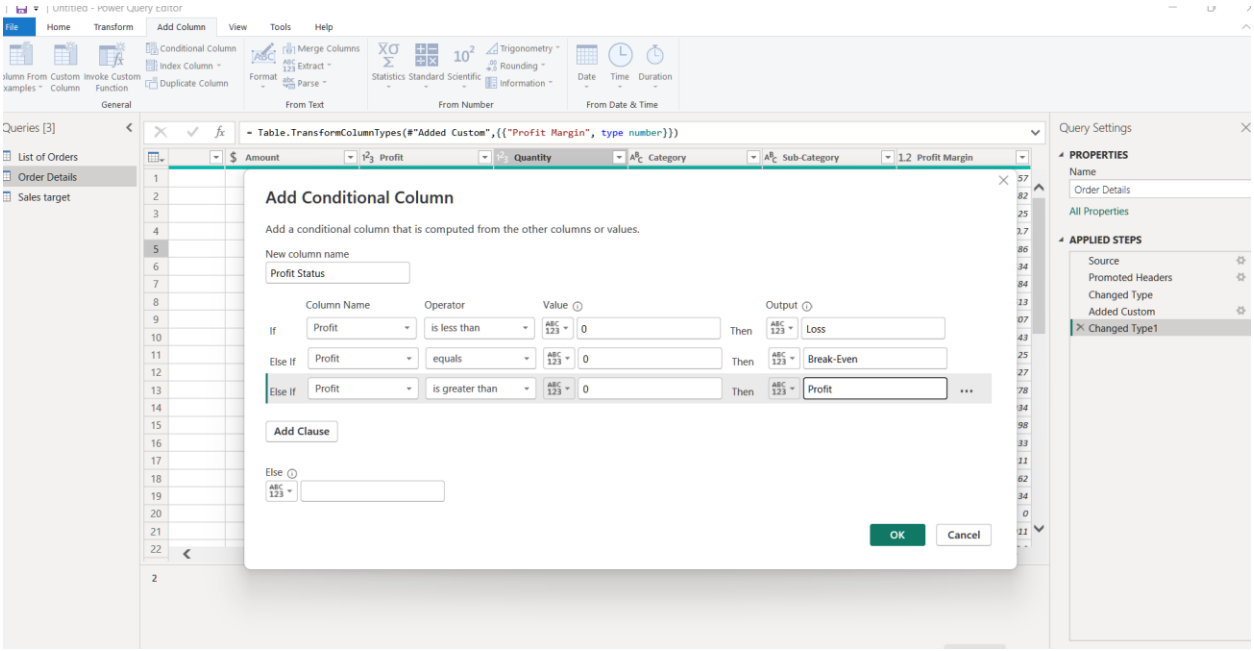


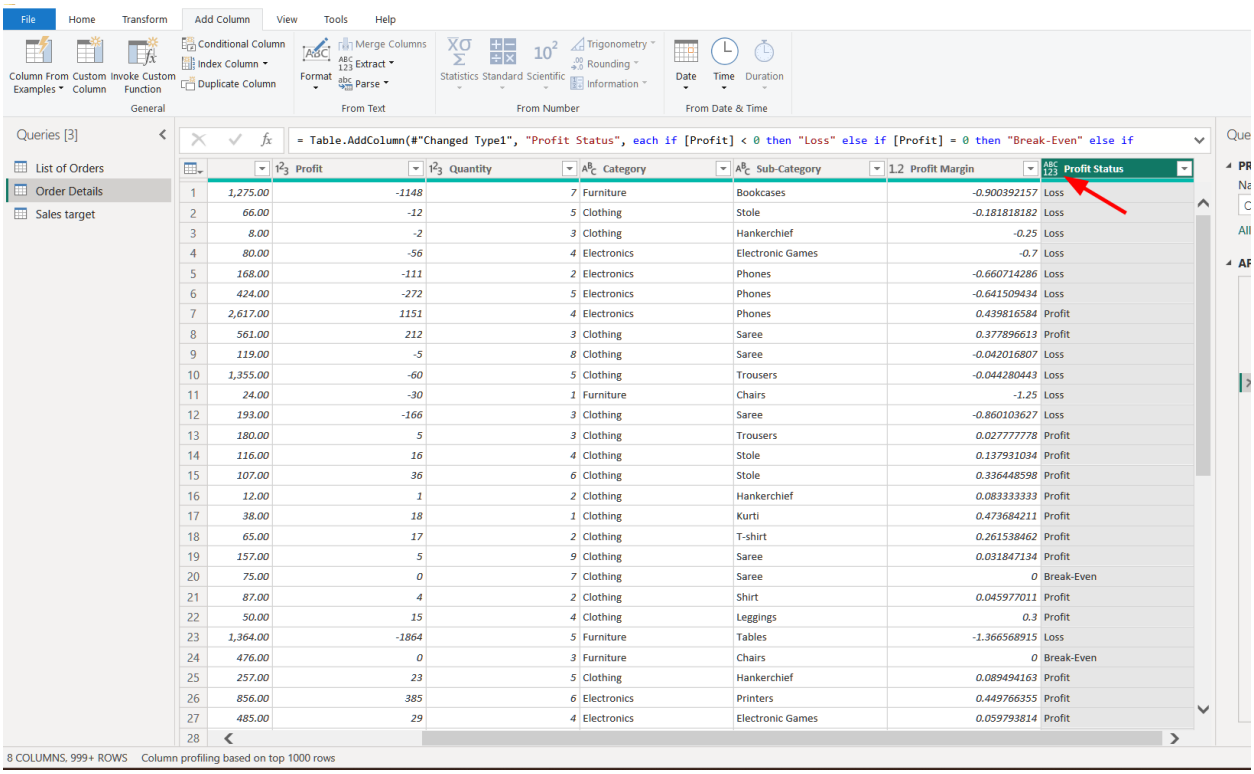
	Order ID	Amount	Profit	Quantity	Category	Sub-Category
1	B-25601					
2	B-25601					
3	B-25601					
4	B-25601					
5	B-25602					
6	B-25602					
7	B-25602					
8	B-25602					
9	B-25602					
10	B-25603					
11	B-25603					
12	B-25603					
13	B-25603					
14	B-25603					
15	B-25603					
16	B-25603					
17	B-25603					
18	B-25604					
19	B-25604					
20	B-25605					
21	B-25606	87.00	4	2	Clothing	Shirt
22	B-25607	50.00	15	4	Clothing	Leggings
23	B-25608	1,364.00	-1864	5	Furniture	Tables
24	B-25608	476.00	0	3	Furniture	Chairs
25	B-25608	257.00	23	5	Clothing	Hankerchief
26	B-25608	856.00	385	6	Electronics	Printers
27	B-25609	485.00	29	4	Electronics	Electronic Games
28	B-25609	25.00	-5	4	Clothing	Saree



Conditional Column:

Added a conditional column named Profit Status using conditional logic in Power Query Editor. Classified records as Loss, Break-Even, or Profit based on profit values to support profitability analysis.

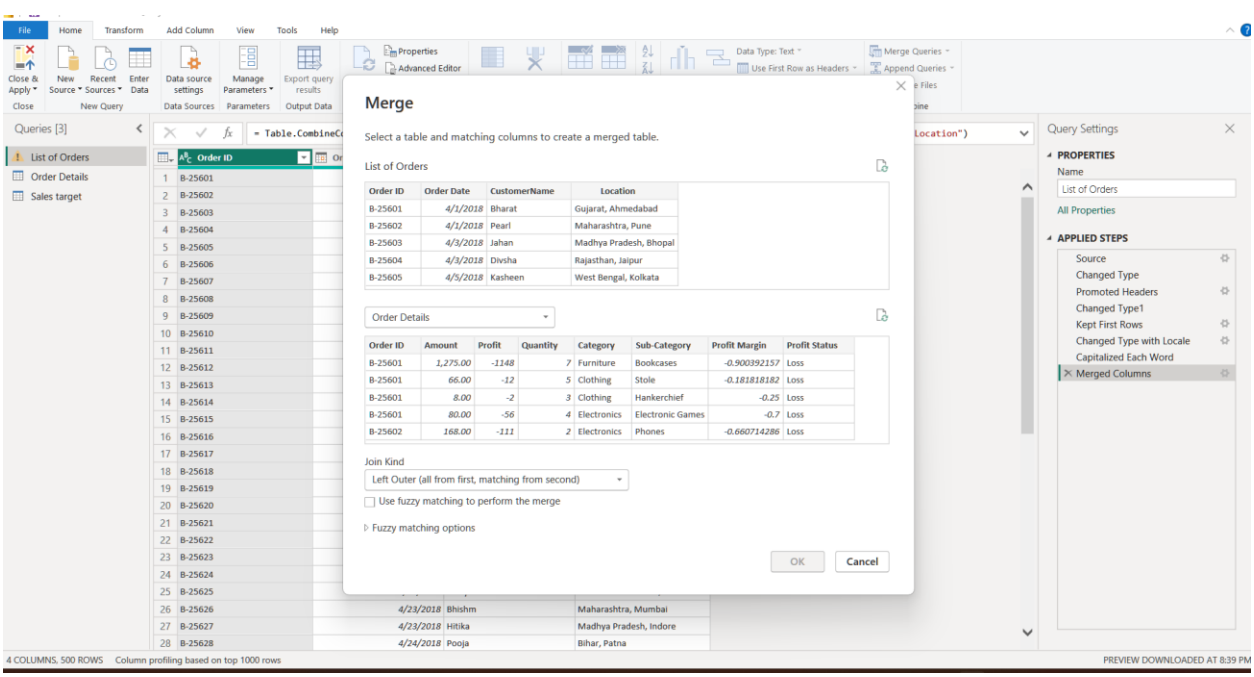


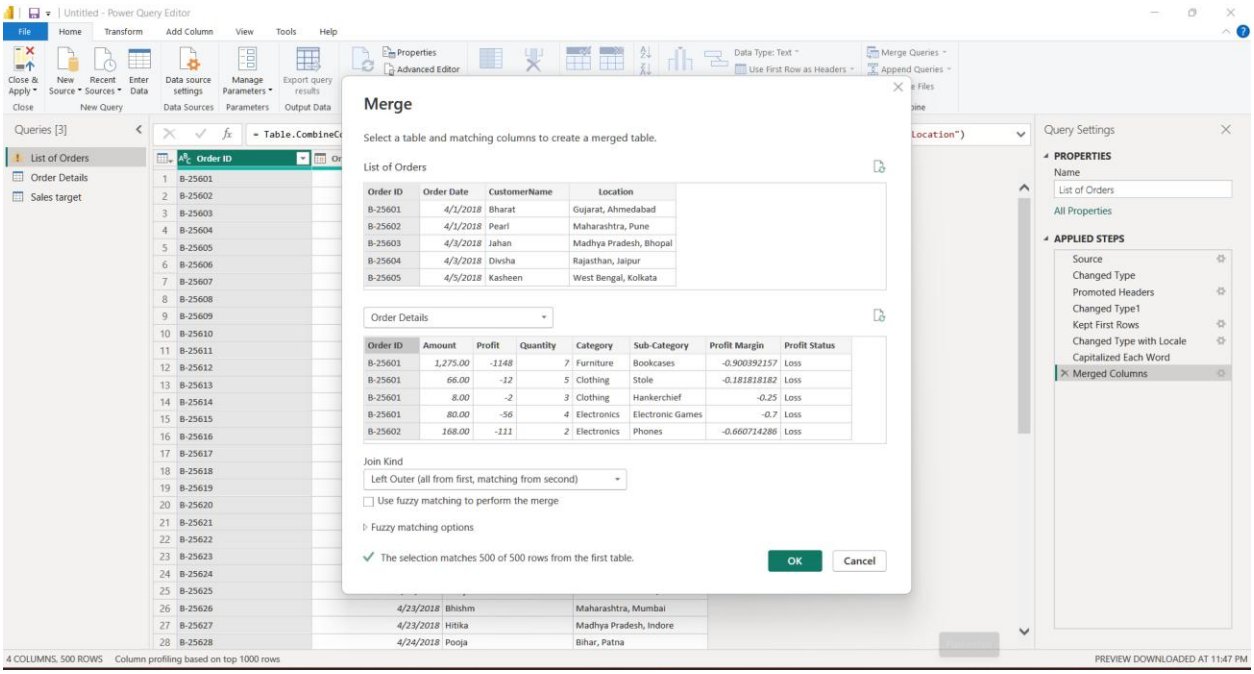


	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1	1,275.00	-1148	7	Furniture	Bookcases	-0.900392157	Loss
2	66.00	-12	5	Clothing	Stole	-0.181818182	Loss
3	8.00	-2	3	Clothing	Hankerchief	-0.25	Loss
4	80.00	-56	4	Electronics	Electronic Games	-0.7	Loss
5	168.00	-111	2	Electronics	Phones	-0.660714286	Loss
6	424.00	-272	5	Electronics	Phones	-0.641509434	Loss
7	2,617.00	1151	4	Electronics	Phones	0.439816584	Profit
8	561.00	212	3	Clothing	Saree	0.377896613	Profit
9	119.00	-5	8	Clothing	Saree	-0.042016807	Loss
10	1,355.00	-60	5	Clothing	Trousers	-0.044280443	Loss
11	24.00	-30	1	Furniture	Chairs	-1.25	Loss
12	193.00	-166	3	Clothing	Saree	-0.860103627	Loss
13	180.00	5	3	Clothing	Trousers	0.027777778	Profit
14	116.00	16	4	Clothing	Stole	0.137931034	Profit
15	107.00	36	6	Clothing	Stole	0.336448598	Profit
16	12.00	1	2	Clothing	Hankerchief	0.083333333	Profit
17	38.00	18	1	Clothing	Kurti	0.473684211	Profit
18	65.00	17	2	Clothing	T-shirt	0.261538462	Profit
19	157.00	5	9	Clothing	Saree	0.031847134	Profit
20	75.00	0	7	Clothing	Saree	0	Break-Even
21	87.00	4	2	Clothing	Shirt	0.045977011	Profit
22	50.00	15	4	Clothing	Leggings	0.3	Profit
23	1,364.00	-1864	5	Furniture	Tables	-1.366568915	Loss
24	476.00	0	3	Furniture	Chairs	0	Break-Even
25	257.00	23	5	Clothing	Hankerchief	0.089494163	Profit
26	856.00	385	6	Electronics	Printers	0.449766355	Profit
27	485.00	29	4	Electronics	Electronic Games	0.059793814	Profit

Merging Data (Joins):

Merged the List of Orders and Order Details tables using the Order ID field with a Left Outer Join in Power Query Editor. Expanded the required columns and renamed the merged table as Orders Data for unified analysis.




Untitled - Power Query Editor
Merge
Select a table and matching columns to create a merged table.
List of Orders
Order ID | Order Date | CustomerName | Location
B-25601 | 4/1/2018 | Bharat | Gujarat, Ahmedabad
B-25602 | 4/1/2018 | Pearl | Maharashtra, Pune
B-25603 | 4/3/2018 | Jahan | Madhya Pradesh, Bhopal
B-25604 | 4/3/2018 | Divsha | Rajasthan, Jaipur
B-25605 | 4/5/2018 | Kasheen | West Bengal, Kolkata
Order Details
Order ID | Amount | Profit | Quantity | Category | Sub-Category | Profit Margin | Profit Status
B-25601 | 1,275.00 | -1148 | 7 | Furniture | Bookcases | -0.900392157 | Loss
B-25601 | 66.00 | -12 | 5 | Clothing | Stole | -0.181818182 | Loss
B-25601 | 8.00 | -2 | 3 | Clothing | Hankerchief | -0.25 | Loss
B-25601 | 80.00 | -56 | 4 | Electronics | Electronic Games | -0.7 | Loss
B-25602 | 168.00 | -111 | 2 | Electronics | Phones | -0.660714286 | Loss
Join Kind
Left Outer (all from first, matching from second)
Use fuzzy matching to perform the merge
Fuzzy matching options
The selection matches 500 of 500 rows from the first table.
OK
Cancel
Queries [3]
List of Orders
Order Details
Sales target
Query Settings
PROPERTIES
Name
List of Orders
All Properties
APPLIED STEPS
Source
Changed Type
Promoted Headers
Changed Type1
Kept First Rows
Changed Type with Locale
Capitalized Each Word
Merged Columns
4 COLUMNS, 500 ROWS
Column profiling based on top 1000 rows
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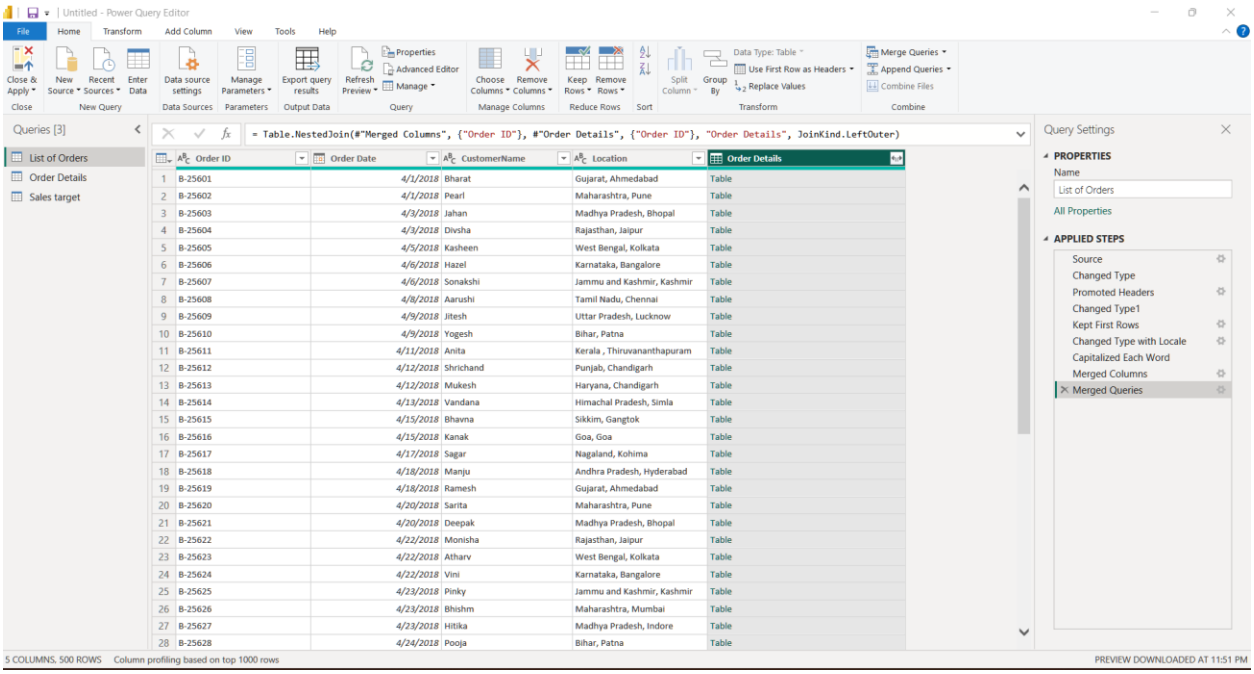
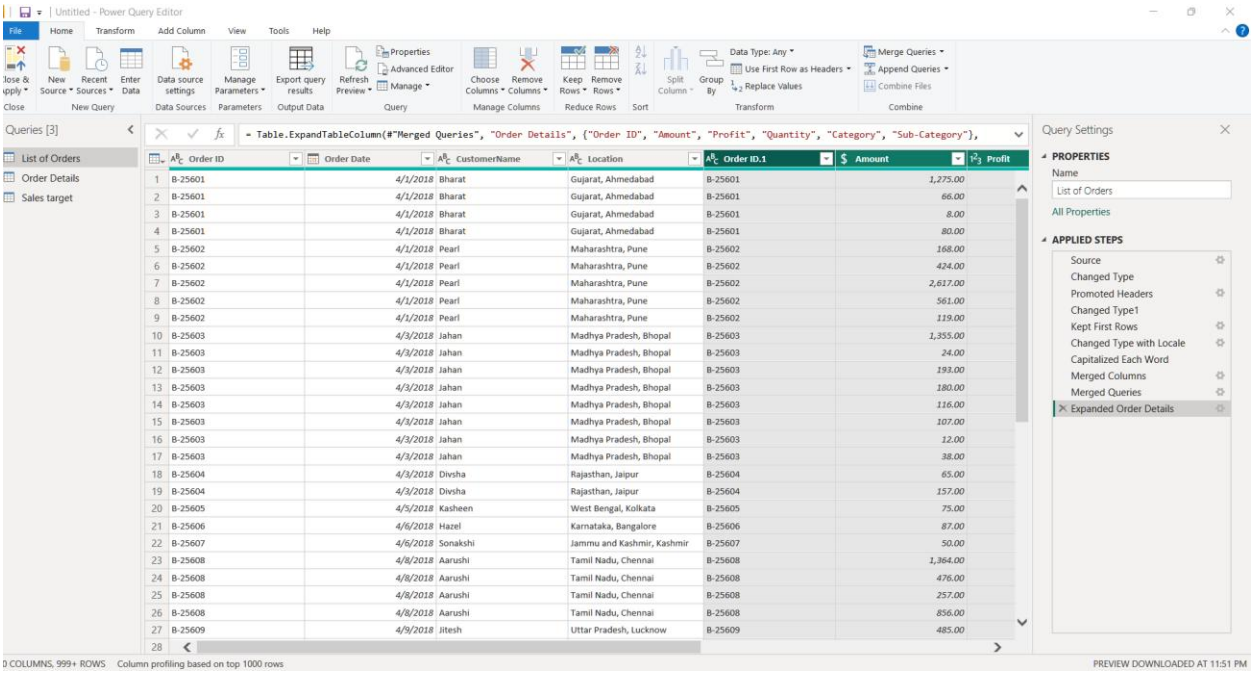

Untitled - Power Query Editor
= Table.NestedJoin(#"Merged Columns", {"Order ID"}, #"Order Details", {"Order ID"}, "Order Details", JoinKind.LeftOuter)
Order ID | Order Date | CustomerName | Location | Order Details
1 B-25601 4/1/2018 Bharat Gujarat, Ahmedabad Table
2 B-25602 4/1/2018 Pearl Maharashtra, Pune Table
3 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal Table
4 B-25604 4/3/2018 Divsha Rajasthan, Jaipur Table
5 B-25605 4/5/2018 Kasheen West Bengal, Kolkata Table
6 B-25606 4/6/2018 Hazel Karnataka, Bangalore Table
7 B-25607 4/6/2018 Sonakshi Jammu and Kashmir, Kashmir Table
8 B-25608 4/8/2018 Aarushi Tamil Nadu, Chennai Table
9 B-25609 4/9/2018 Jitesh Uttar Pradesh, Lucknow Table
10 B-25610 4/9/2018 Yogesh Bihar, Patna Table
11 B-25611 4/11/2018 Anita Kerala , Thiruvananthapuram Table
12 B-25612 4/12/2018 Shrichand Punjab, Chandigarh Table
13 B-25613 4/12/2018 Mukesh Haryana, Chandigarh Table
14 B-25614 4/13/2018 Vandana Himachal Pradesh, Simla Table
15 B-25615 4/15/2018 Bhavna Sikkim, Gangtok Table
16 B-25616 4/15/2018 Kanak Goa, Goa Table
17 B-25617 4/17/2018 Sagar Nagaland, Kohima Table
18 B-25618 4/18/2018 Manju Andhra Pradesh, Hyderabad Table
19 B-25619 4/18/2018 Ramesh Gujarat, Ahmedabad Table
20 B-25620 4/20/2018 Sarita Maharashtra, Pune Table
21 B-25621 4/20/2018 Deepak Madhya Pradesh, Bhopal Table
22 B-25622 4/22/2018 Monisha Rajasthan, Jaipur Table
23 B-25623 4/22/2018 Atharv West Bengal, Kolkata Table
24 B-25624 4/22/2018 Vini Karnataka, Bangalore Table
25 B-25625 4/23/2018 Pinky Jammu and Kashmir, Kashmir Table
26 B-25626 4/23/2018 Bhishm Maharashtra, Mumbai Table
27 B-25627 4/23/2018 Hitika Madhya Pradesh, Indore Table
28 B-25628 4/24/2018 Pooja Bihar, Patna Table
Queries [3]
List of Orders
Order Details
Sales target
Query Settings
PROPERTIES
Name
List of Orders
All Properties
APPLIED STEPS
Source
Changed Type
Promoted Headers
Changed Type1
Kept First Rows
Changed Type with Locale
Capitalized Each Word
Merged Columns
Merged Queries
5 COLUMNS, 500 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 11:51 PM



Untitled - Power Query Editor
File
Home
Transform
Add Column
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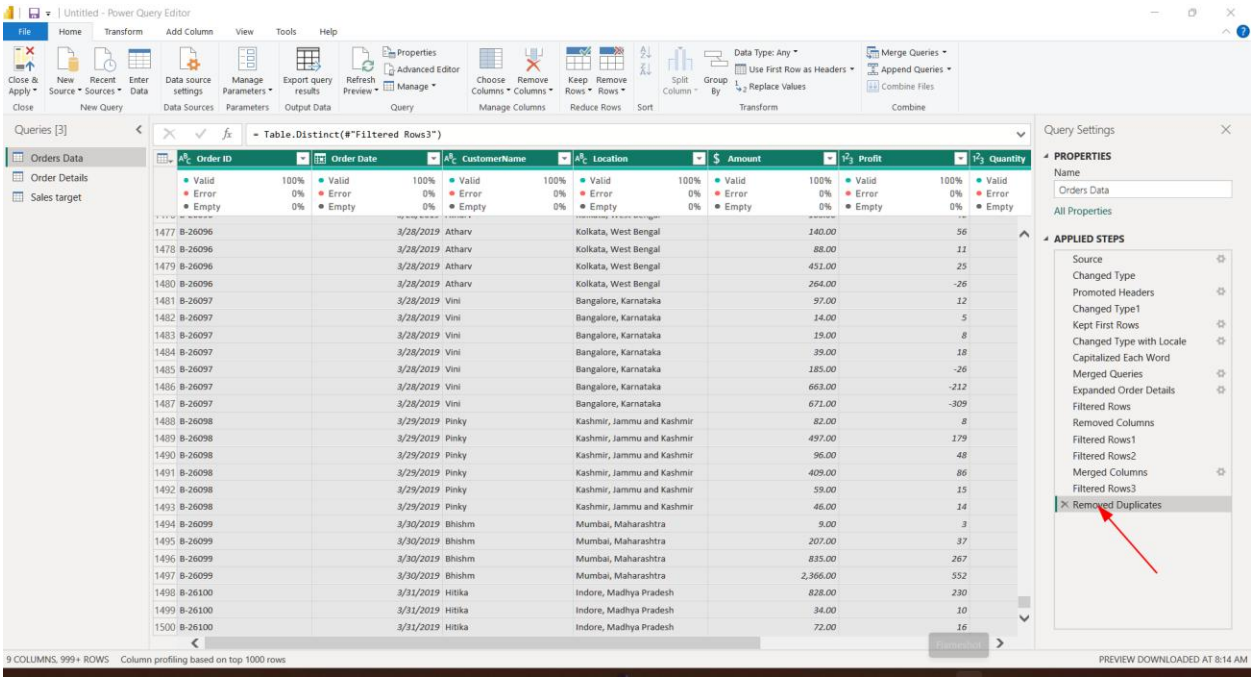
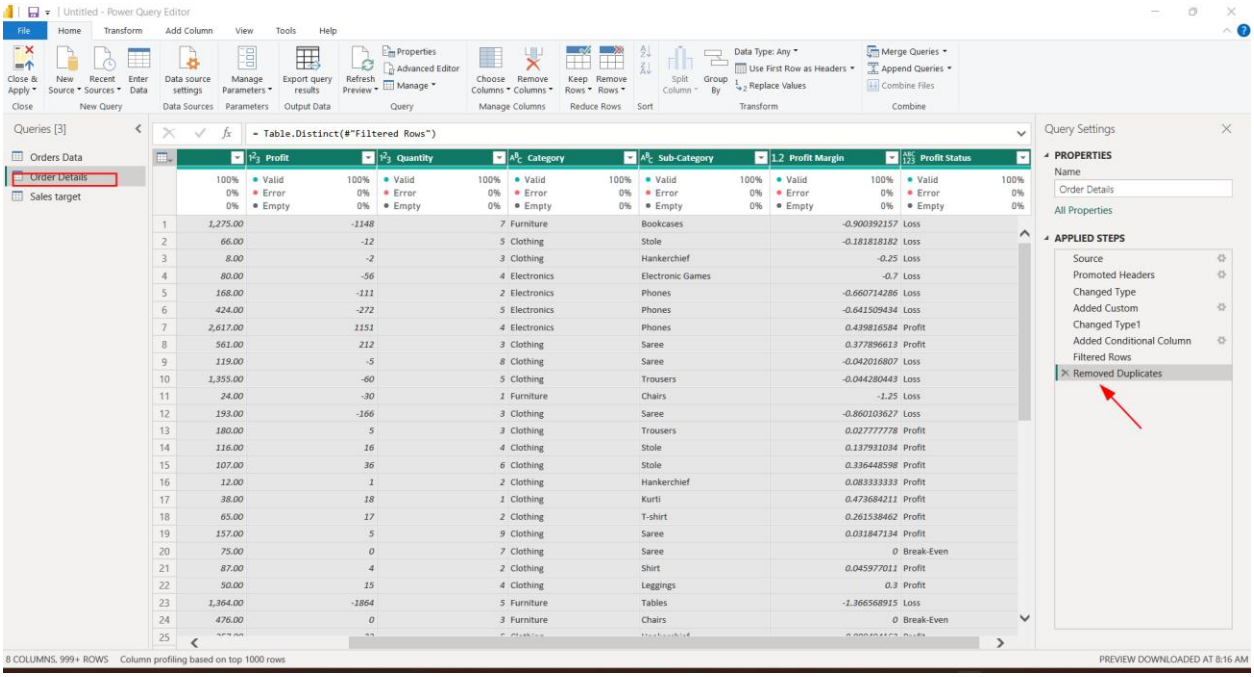
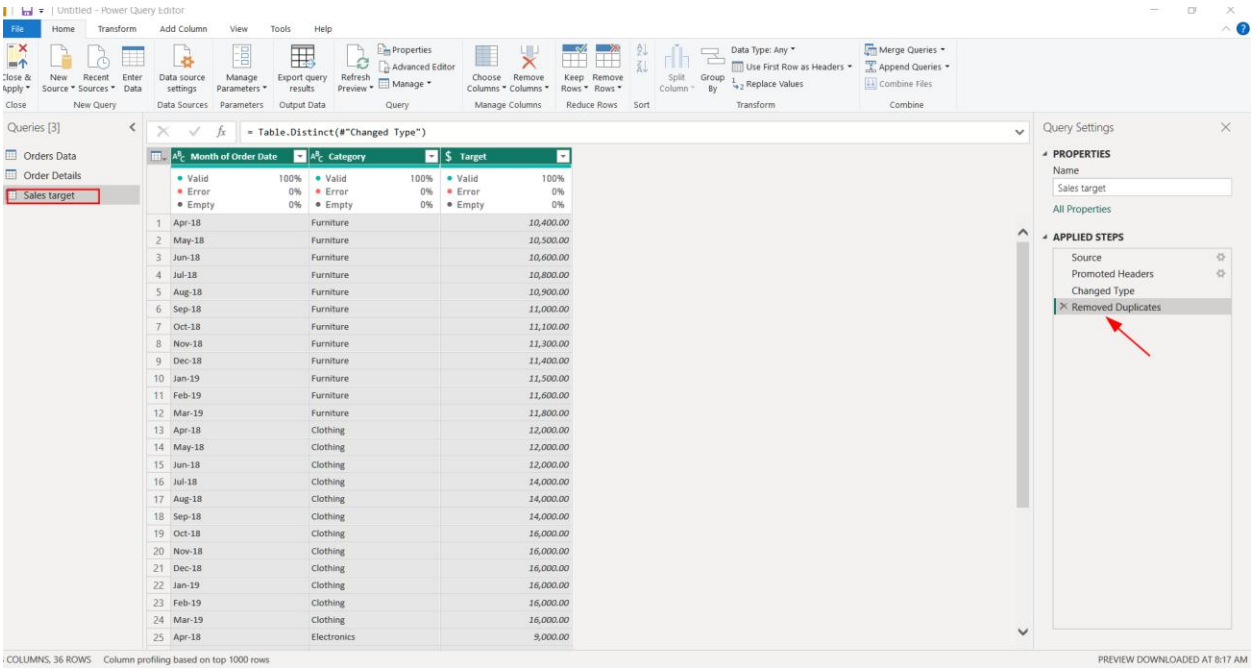
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Export query results
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Any
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Close
New Query
Data Sources
Parameters
Output Data
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
Queries [3]
List of Orders
Order Details
Sales target
= Table.ExpandTableColumn(#"Merged Queries", "Order Details", {"Order ID", "Amount", "Profit", "Quantity", "Category", "Sub-Category"},
Order ID
Order Date
CustomerName
Location
Order ID.1
Amount
Profit
1 B-25601 4/1/2018 Bharat Gujarat, Ahmedabad B-25601 1,275.00
2 B-25601 4/1/2018 Bharat Gujarat, Ahmedabad B-25601 66.00
3 B-25601 4/1/2018 Bharat Gujarat, Ahmedabad B-25601 8.00
4 B-25601 4/1/2018 Bharat Gujarat, Ahmedabad B-25601 80.00
5 B-25602 4/1/2018 Pearl Maharashtra, Pune B-25602 168.00
6 B-25602 4/1/2018 Pearl Maharashtra, Pune B-25602 424.00
7 B-25602 4/1/2018 Pearl Maharashtra, Pune B-25602 2,617.00
8 B-25602 4/1/2018 Pearl Maharashtra, Pune B-25602 561.00
9 B-25602 4/1/2018 Pearl Maharashtra, Pune B-25602 119.00
10 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 1,355.00
11 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 24.00
12 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 193.00
13 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 180.00
14 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 116.00
15 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 107.00
16 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 12.00
17 B-25603 4/3/2018 Jahan Madhya Pradesh, Bhopal B-25603 38.00
18 B-25604 4/3/2018 Divsha Rajasthan, Jaipur B-25604 65.00
19 B-25604 4/3/2018 Divsha Rajasthan, Jaipur B-25604 157.00
20 B-25605 4/5/2018 Kasheen West Bengal, Kolkata B-25605 75.00
21 B-25606 4/6/2018 Hazel Karnataka, Bangalore B-25606 87.00
22 B-25607 4/6/2018 Sonakshi Jammu and Kashmir, Kashmir B-25607 50.00
23 B-25608 4/8/2018 Aarushi Tamil Nadu, Chennai B-25608 1,364.00
24 B-25608 4/8/2018 Aarushi Tamil Nadu, Chennai B-25608 476.00
25 B-25608 4/8/2018 Aarushi Tamil Nadu, Chennai B-25608 257.00
26 B-25608 4/8/2018 Aarushi Tamil Nadu, Chennai B-25608 856.00
27 B-25609 4/9/2018 Jitesh Uttar Pradesh, Lucknow B-25609 485.00
Query Settings
PROPERTIES
Name
List of Orders
All Properties
APPLIED STEPS
Source
Changed Type
Promoted Headers
Changed Type1
Kept First Rows
Changed Type with Locale
Capitalized Each Word
Merged Columns
Merged Queries
Expanded Order Details
0 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 11:51 PM

Handling Missing Data & Duplicate Data:

Checked all tables for missing values, nulls, and errors using column quality indicators in Power Query Editor. No missing or erroneous values were found in the final Orders Data table. Duplicate handling was performed carefully using business logic, retaining repeated Order IDs where multiple products belonged to the same order, and removing only completely identical duplicate records.

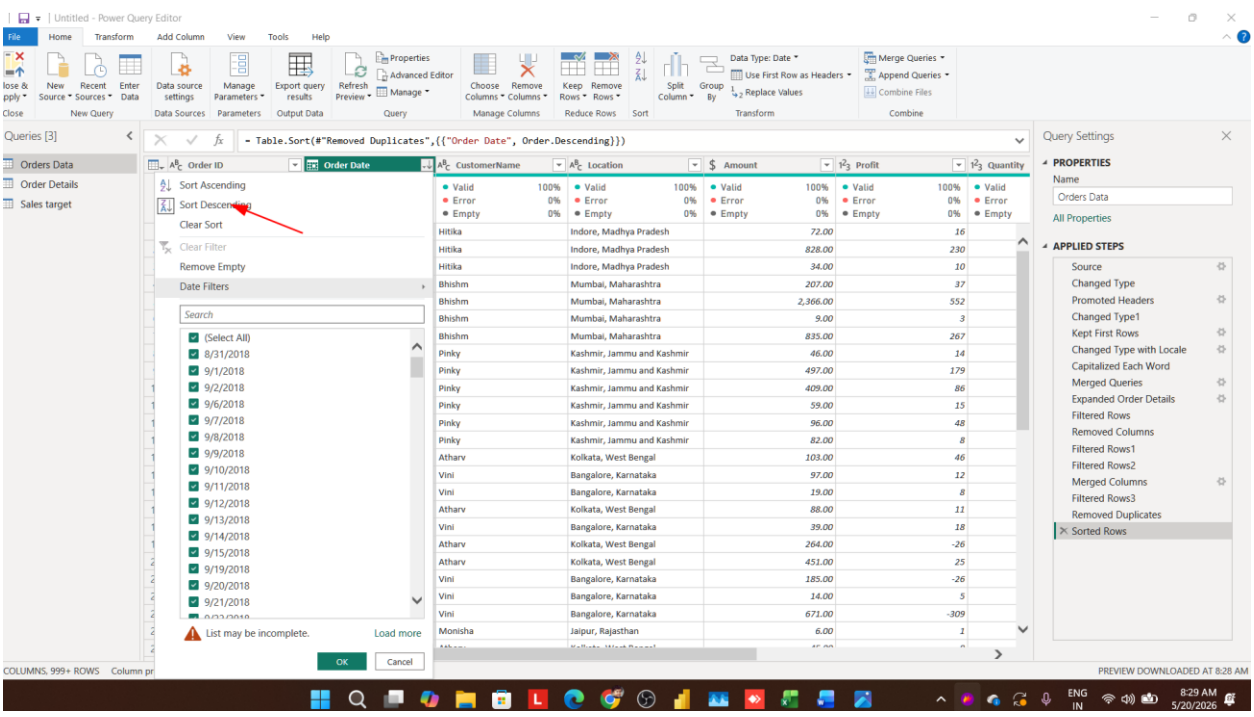


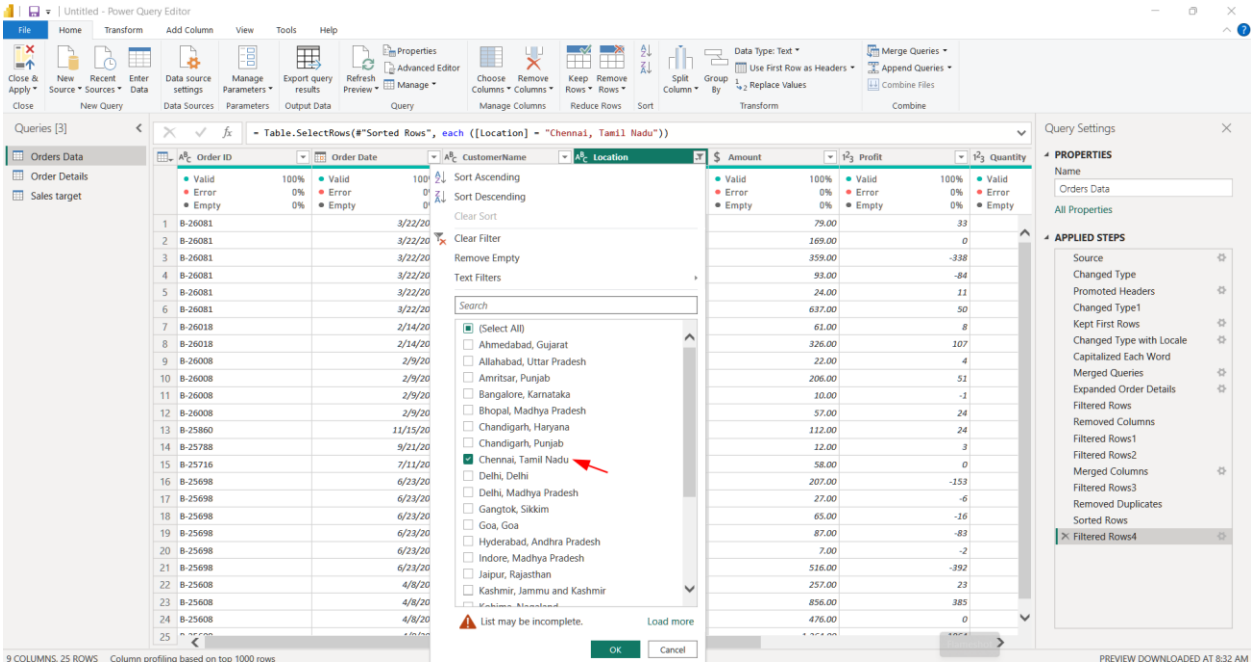




Sorting and Filtering Data:

Sorted the Orders Data table by Order Date in descending order to display the most recent orders first. Applied filters to analyze orders from a specific state, Tamil Nadu, enabling focused regional analysis.

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File
Home
Transform
Add Column
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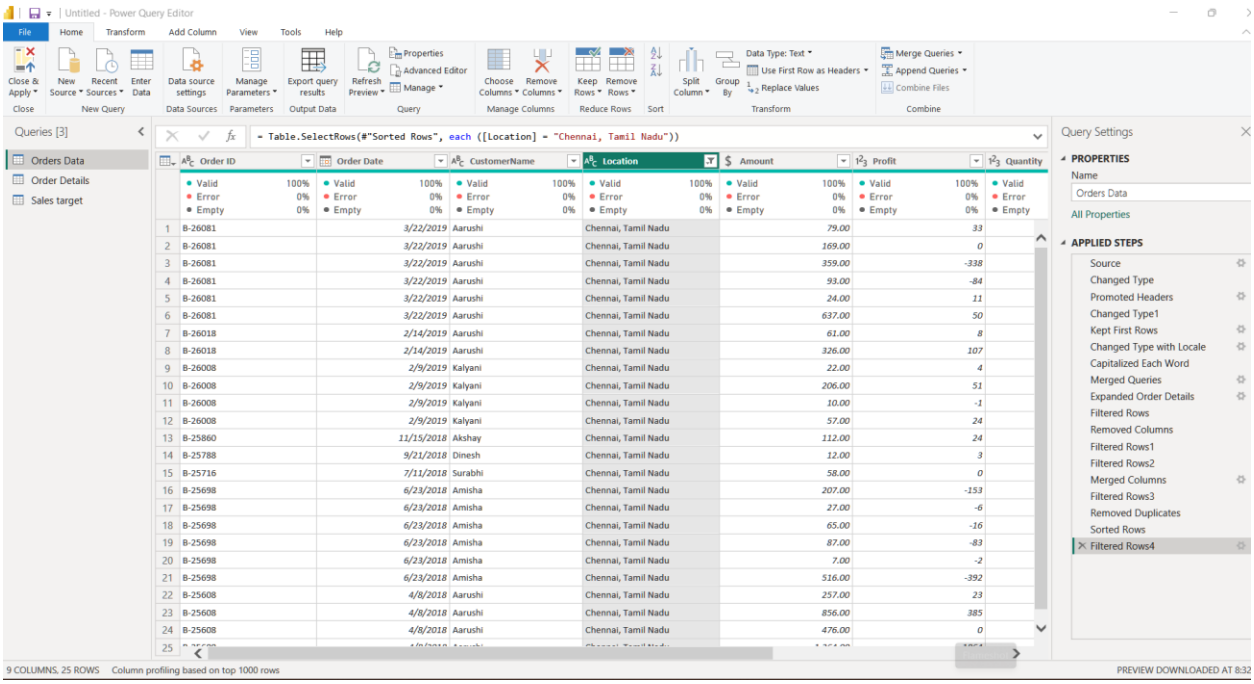
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Export query results
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Text
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Close
New Query
Data Sources
Parameters
Output Data
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
Queries [3]
Orders Data
Order Details
Sales target
= Table.SelectRows(#"Sorted Rows", each ([Location] = "Chennai, Tamil Nadu"))
Order ID
Order Date
CustomerName
Location
Amount
Profit
Quantity
Sort Ascending
Sort Descending
Clear Sort
Clear Filter
Remove Empty
Text Filters
Search
(Select All)
Ahmedabad, Gujarat
Allahabad, Uttar Pradesh
Amritsar, Punjab
Bangalore, Karnataka
Bhopal, Madhya Pradesh
Chandigarh, Haryana
Chandigarh, Punjab
Chennai, Tamil Nadu
Delhi, Delhi
Delhi, Madhya Pradesh
Gangtok, Sikkim
Goa, Goa
Hyderabad, Andhra Pradesh
Indore, Madhya Pradesh
Jaipur, Rajasthan
Kashmir, Jammu and Kashmir
List may be incomplete.
Load more
OK
Cancel
Query Settings
PROPERTIES
Name
Orders Data
All Properties
APPLIED STEPS
Source
Changed Type
Promoted Headers
Changed Type1
Kept First Rows
Changed Type with Locale
Capitalized Each Word
Merged Queries
Expanded Order Details
Filtered Rows
Removed Columns
Filtered Rows1
Filtered Rows2
Merged Columns
Filtered Rows3
Removed Duplicates
Sorted Rows
Filtered Rows4
9 COLUMNS, 25 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 8:32 AM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Close & Apply
New Source
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Enter Data
Data source settings
Manage Parameters
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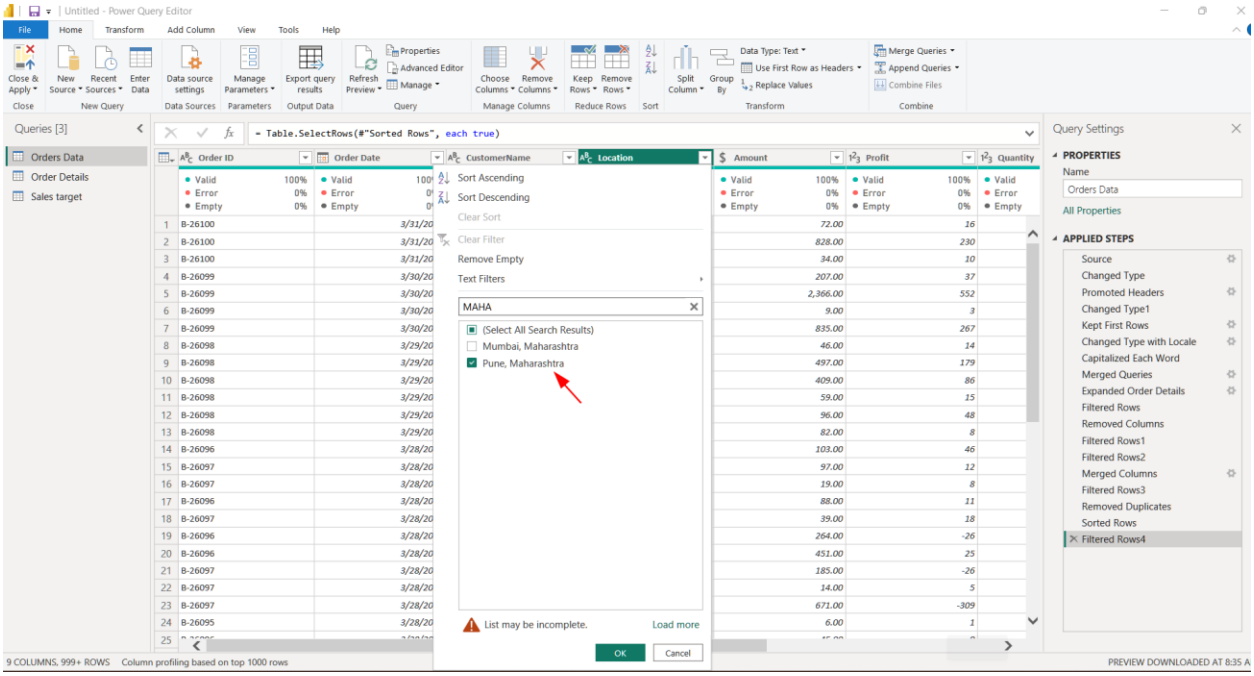
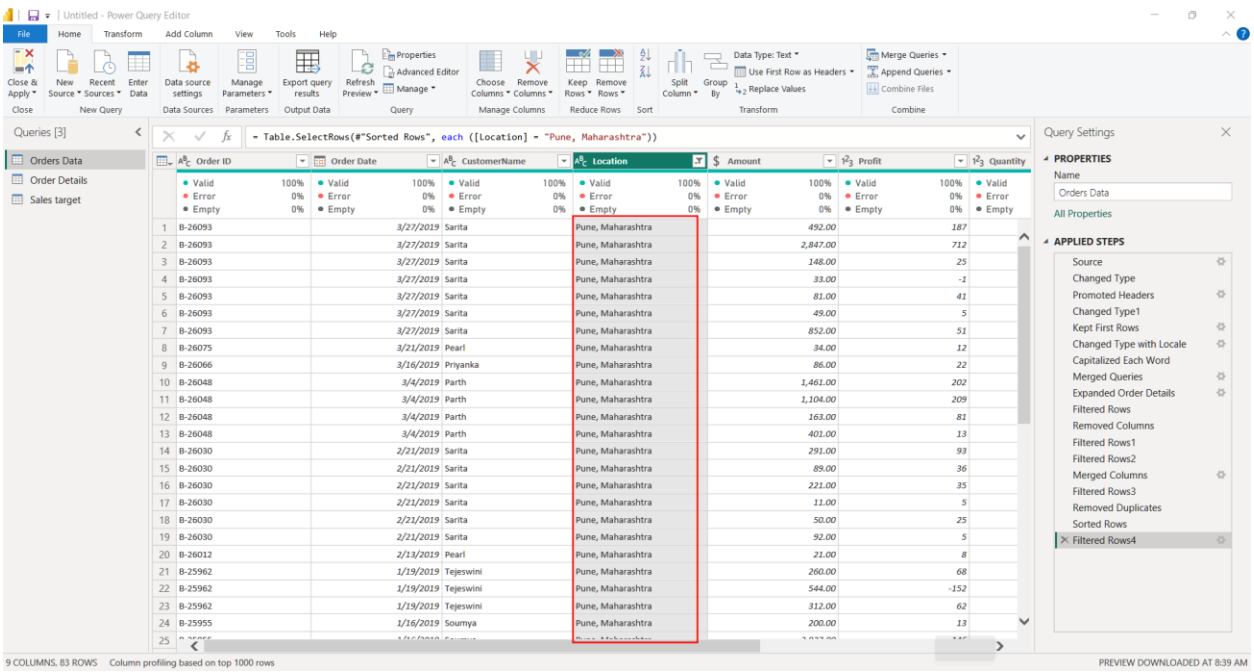
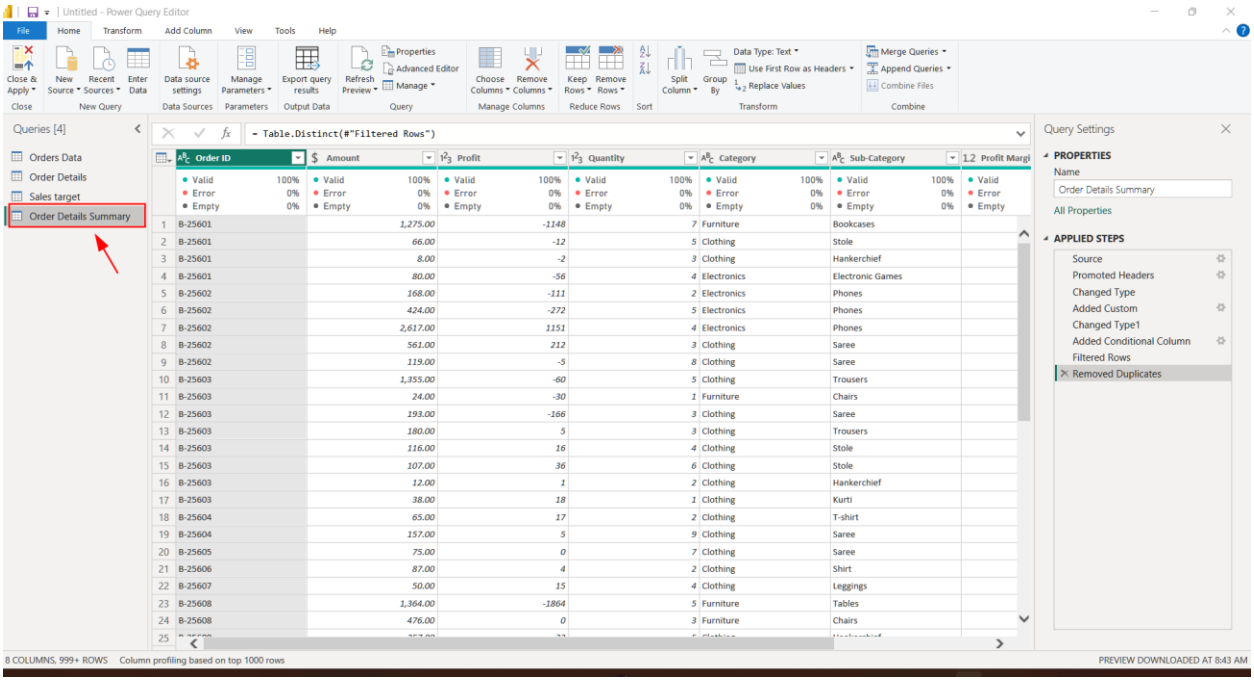
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Text
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Close
New Query
Data Sources
Parameters
Output Data
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
Queries [3]
Orders Data
Order Details
Sales target
= Table.SelectRows(#"Sorted Rows", each ([Location] = "Chennai, Tamil Nadu"))
Order ID
Order Date
CustomerName
Location
Amount
Profit
Quantity
Query Settings
PROPERTIES
Name
Orders Data
All Properties
APPLIED STEPS
Source
Changed Type
Promoted Headers
Changed Type1
Kept First Rows
Changed Type with Locale
Capitalized Each Word
Merged Queries
Expanded Order Details
Filtered Rows
Removed Columns
Filtered Rows1
Filtered Rows2
Merged Columns
Filtered Rows3
Removed Duplicates
Sorted Rows
Filtered Rows4
9 COLUMNS, 25 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 8:32

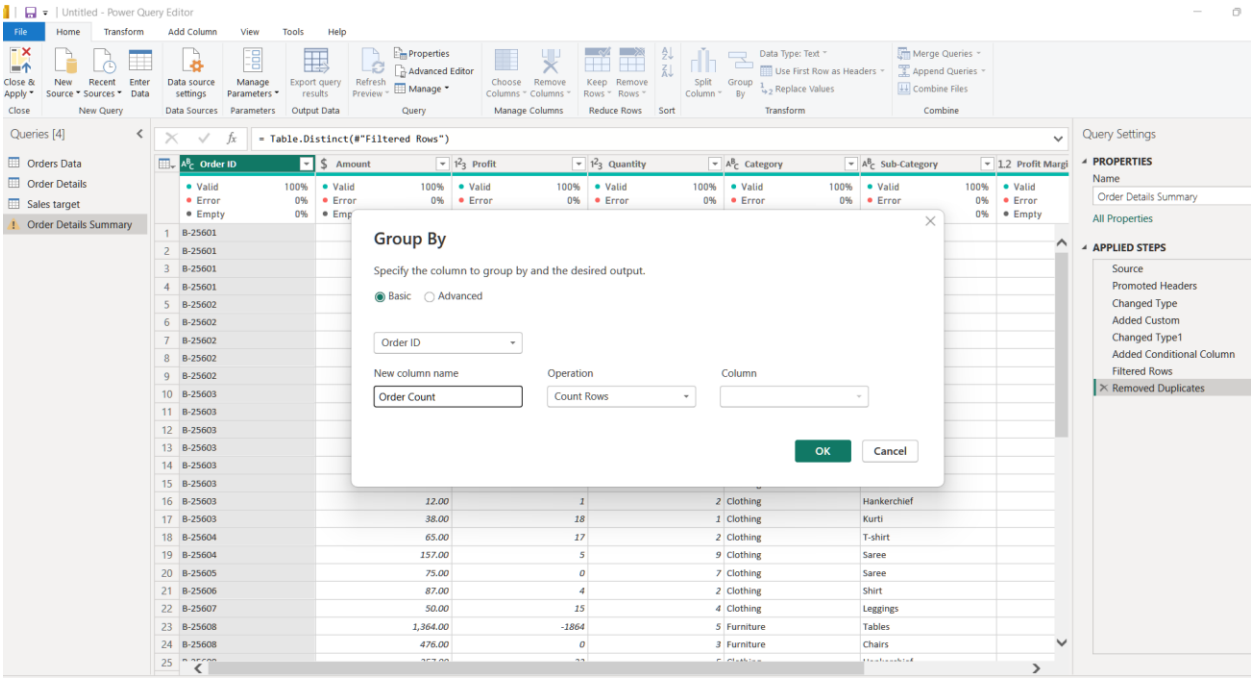


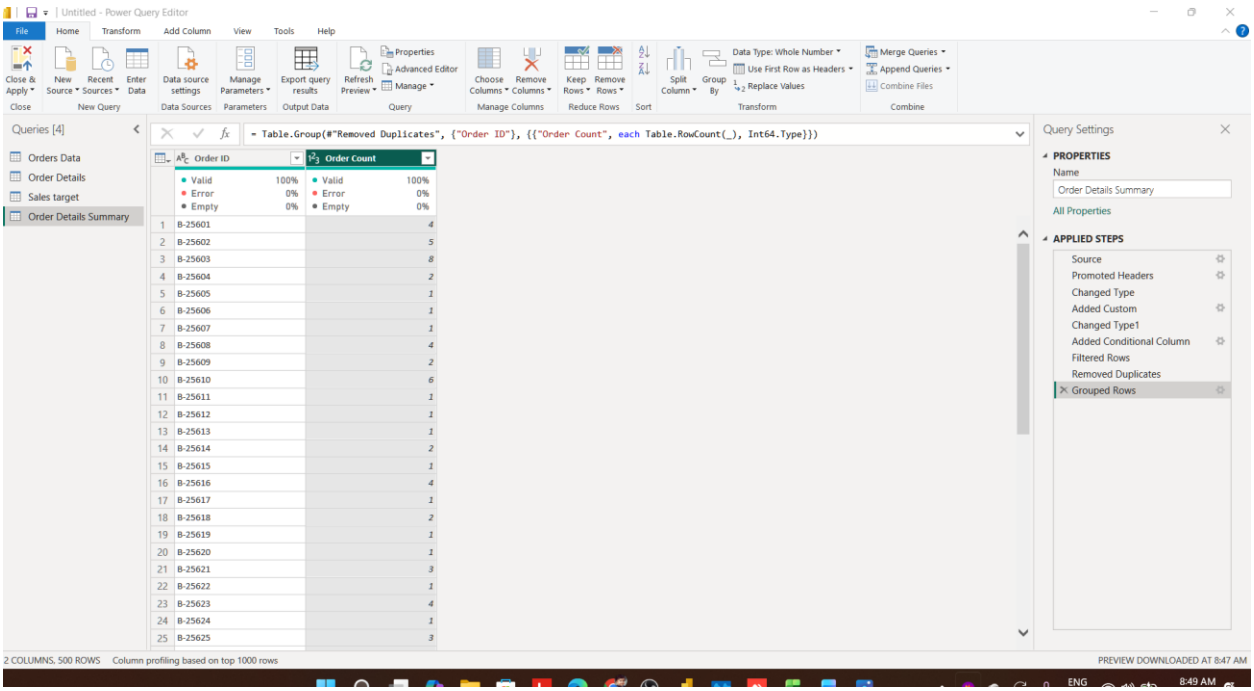


Grouping and Aggregating Data:

Duplicated the Order Details and Sales Target tables to perform aggregation operations. Calculated the count of Order IDs, average profit by category, total amount by sub-category, and total sales target by month using the Group By feature in Power Query Editor for summarized business analysis. Duplicated the Sales Target table and aggregated the Target values by Month of Order Date using the Group By transformation in Power Query Editor to generate monthly sales target summaries.


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [4]
Orders Data
Order Details
Sales target
Order Details Summary
= Table.Distinct(#"Filtered Rows")
Order ID
Amount
Profit
Quantity
Category
Sub-Category
Profit Margi
Query Settings
PROPERTIES
Name
Order Details Summary
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Changed Type1
Added Conditional Column
Filtered Rows
Removed Duplicates
8 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 8:43 AM


Untitled - Power Query Editor
Queries [4]
Orders Data
Order Details
Sales target
Order Details Summary
= Table.Distinct(#"Filtered Rows")
Group By
Specify the column to group by and the desired output.
Basic
Advanced
Order ID
New column name
Order Count
Operation
Count Rows
Column
OK
Cancel
Query Settings
PROPERTIES
Name
Order Details Summary
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Changed Type1
Added Conditional Column
Filtered Rows
Removed Duplicates


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [4]
Orders Data
Order Details
Sales target
Order Details Summary
= Table.Group(#"Removed Duplicates", {"Order ID"}, {{"Order Count", each Table.RowCount(_), Int64.Type}})
Order ID
Order Count
Query Settings
PROPERTIES
Name
Order Details Summary
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Changed Type1
Added Conditional Column
Filtered Rows
Removed Duplicates
Grouped Rows
2 COLUMNS, 500 ROWS
Column profiling based on top 1000 rows
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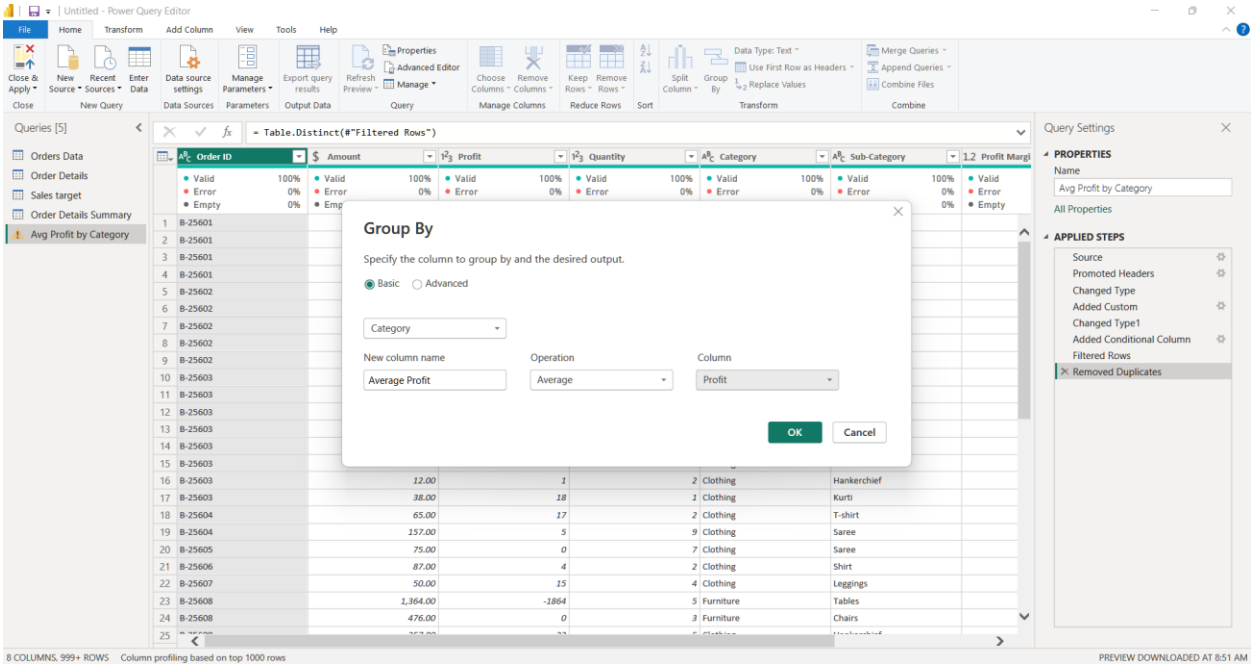
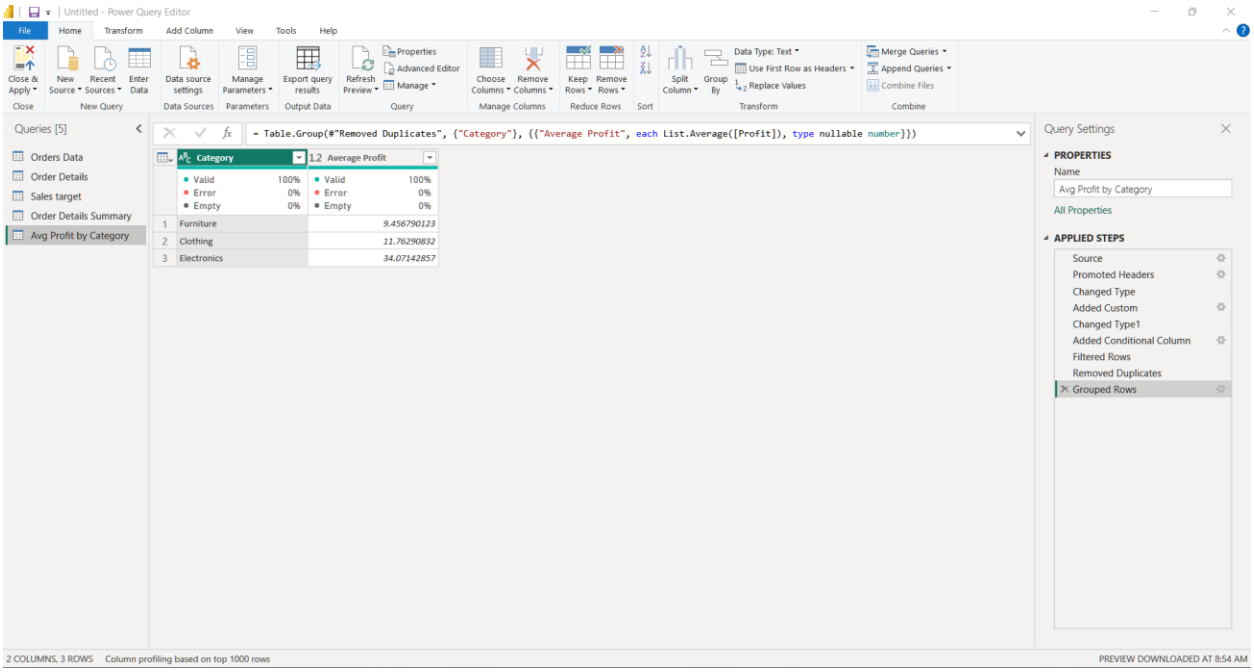

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [5]
Orders Data
Order Details
Sales target
Order Details Summary
Avg Profit by Category
= Table.Distinct(#"Filtered Rows")
Order ID
Amount
Profit
Quantity
Category
Sub-Category
Group By
Specify the column to group by and the desired output.
Basic
Advanced
Category
New column name
Average Profit
Operation
Average
Column
Profit
OK
Cancel
Query Settings
PROPERTIES
Name
Avg Profit by Category
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Changed Type1
Added Conditional Column
Filtered Rows
Removed Duplicates
8 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 8:51 AM


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [5]
Orders Data
Order Details
Sales target
Order Details Summary
Avg Profit by Category
= Table.Group(#"Removed Duplicates", {"Category"}, {{"Average Profit", each List.Average([Profit]), type nullable number}})
Category
Average Profit
Furniture 9.456790123
Clothing 11.76290832
Electronics 34.07142857
Query Settings
PROPERTIES
Name
Avg Profit by Category
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Changed Type1
Added Conditional Column
Filtered Rows
Removed Duplicates
Grouped Rows
2 COLUMNS, 3 ROWS
Column profiling based on top 1000 rows
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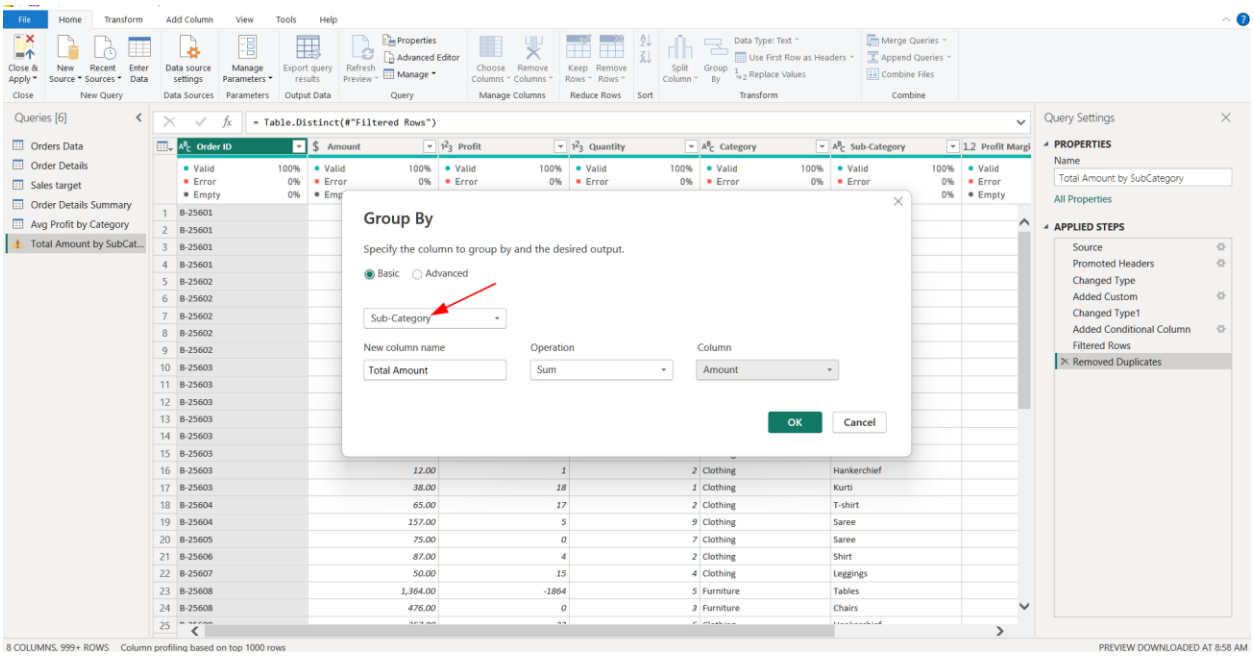
Queries [6]
Orders Data
Order Details
Sales target
Order Details Summary
Avg Profit by Category
Total Amount by SubCat...
= Table.Distinct(#"Filtered Rows")
Order ID
Amount
Profit
Quantity
Category
Sub-Category
Group By
Specify the column to group by and the desired output.
Basic
Advanced
Sub-Category
New column name
Total Amount
Operation
Sum
Column
Amount
OK
Cancel
Query Settings
PROPERTIES
Name
Total Amount by SubCategory
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Added Custom
Changed Type1
Added Conditional Column
Filtered Rows
Removed Duplicates
8 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
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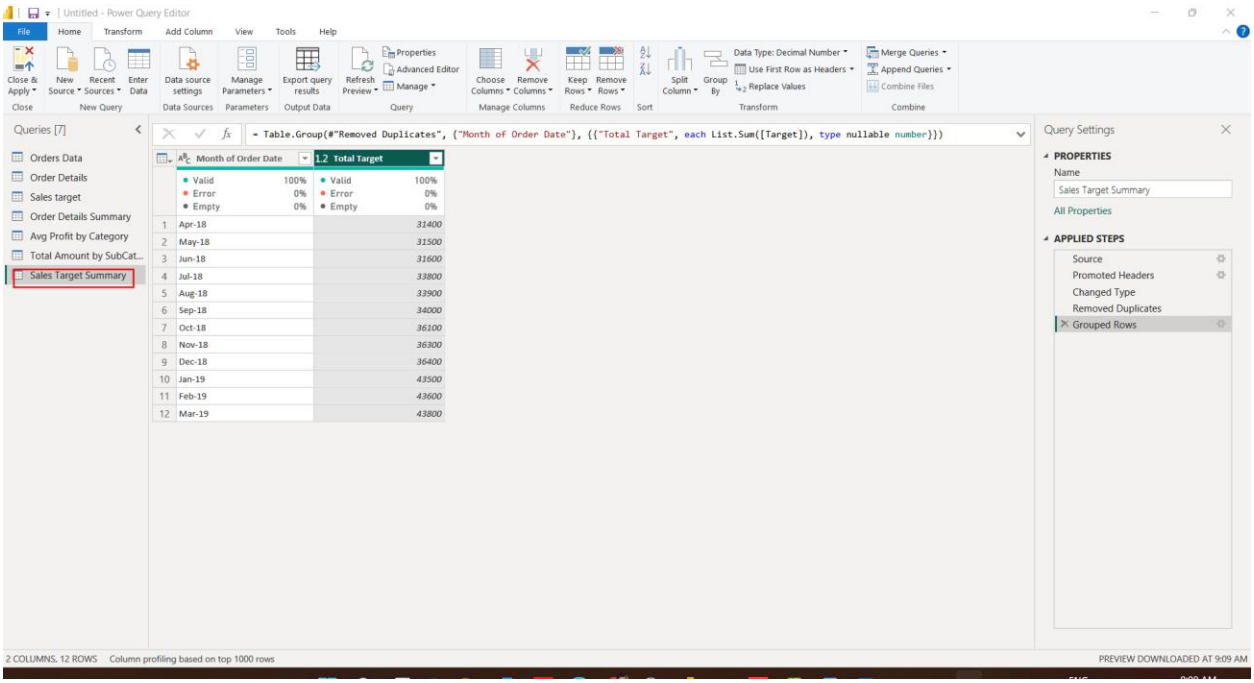
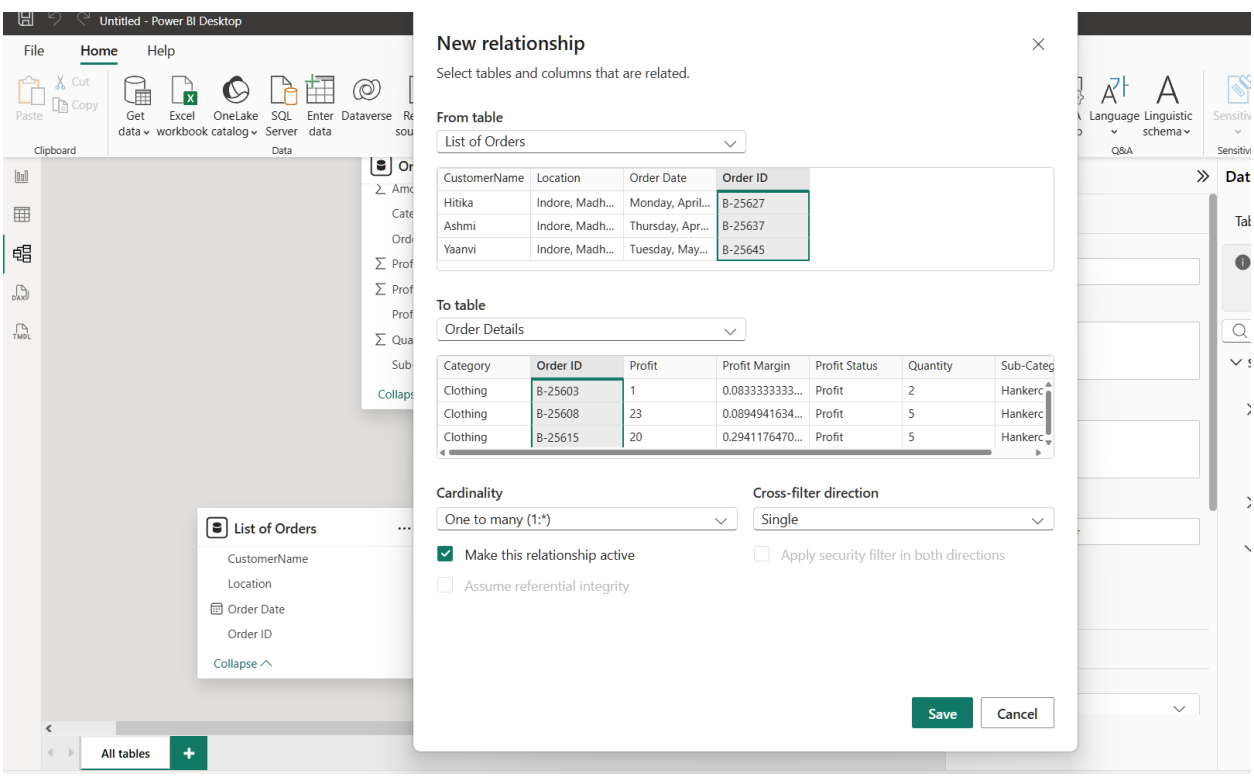
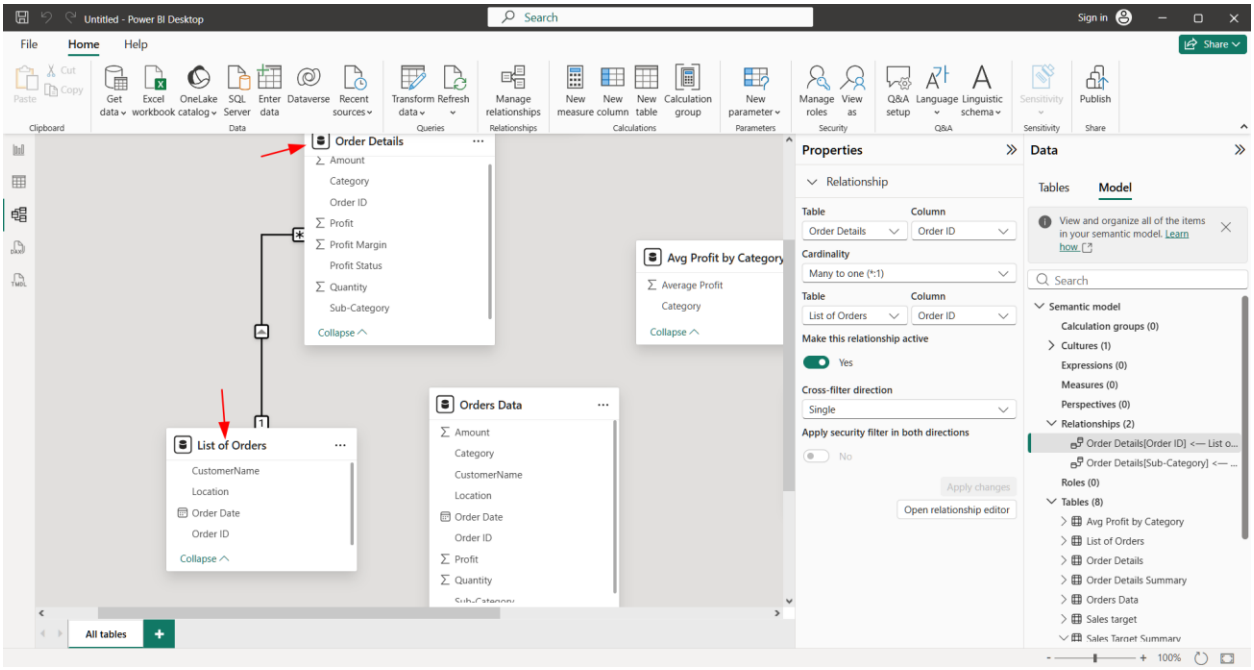
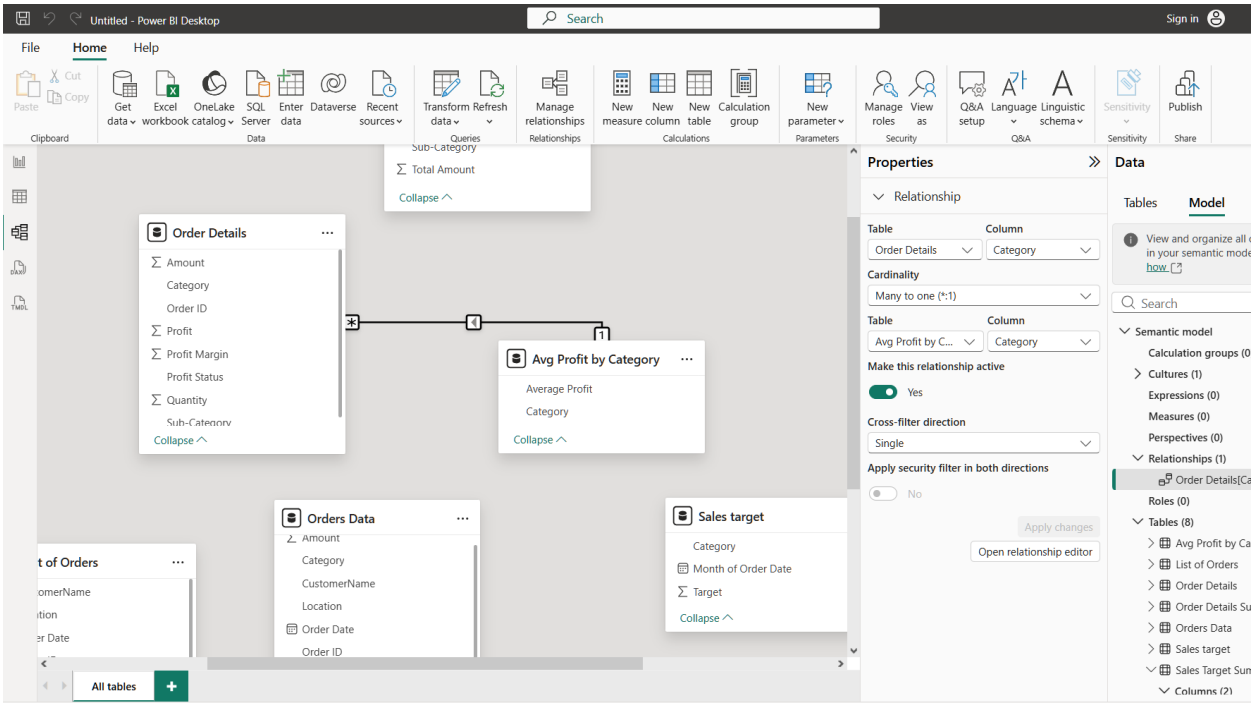


Data Modeling:

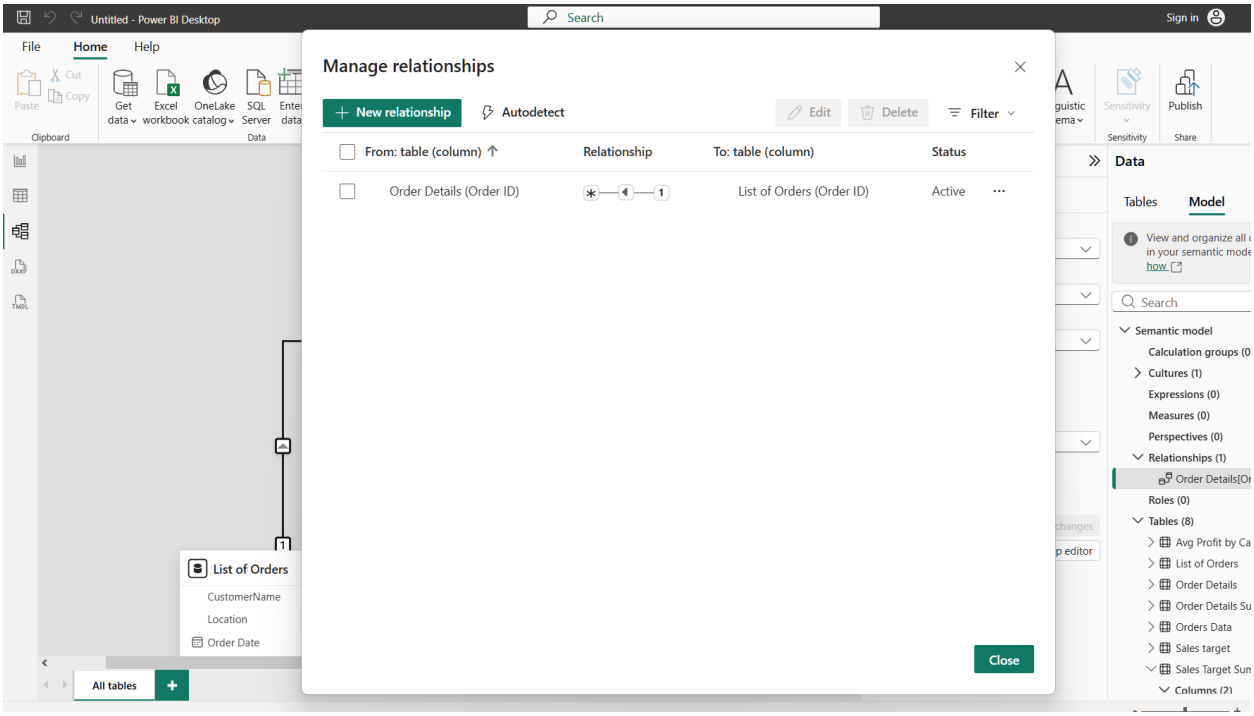
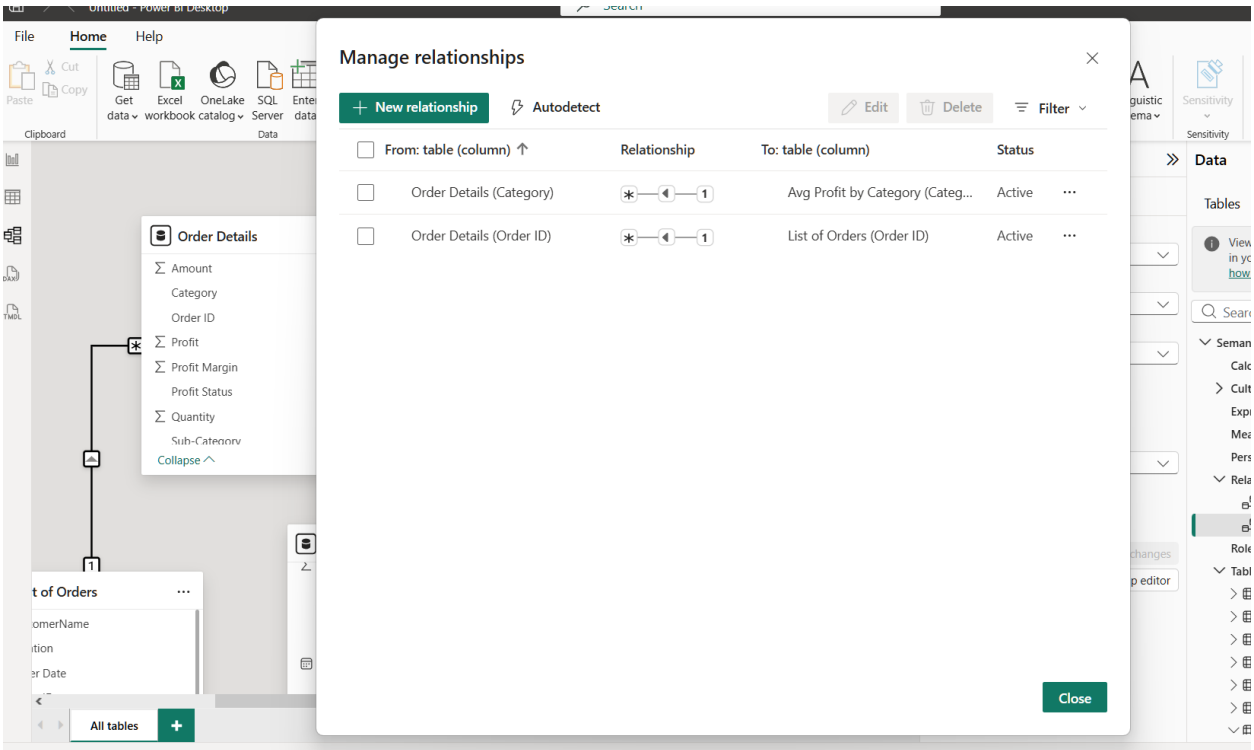
Established data model relationships between List of Orders and Order Details using Order ID, and between Order Details and Sales Target using Category. Verified and managed relationships using the Manage Relationships feature to ensure all relationships were active and correctly configured for accurate reporting and analysis.




Untitled - Power BI Desktop
Search
Sign in
File
Home
Help
Share
Order Details
Amount
Category
Order ID
Profit
Profit Margin
Profit Status
Quantity
Sub-Category
Collapse
Avg Profit by Category
Average Profit
Category
Collapse
Orders Data
Amount
Category
CustomerName
Location
Order Date
Order ID
Profit
Quantity
List of Orders
CustomerName
Location
Order Date
Order ID
Collapse
All tables
Properties
Relationship
Table
Column
Order Details
Order ID
Cardinality
Many to one (*:1)
Table
Column
List of Orders
Order ID
Make this relationship active
Yes
Cross-filter direction
Single
Apply security filter in both directions
No
Apply changes
Open relationship editor
Data
Tables
Model
View and organize all of the items in your semantic model. Learn how
Search
Semantic model
Calculation groups (0)
Cultures (1)
Expressions (0)
Measures (0)
Perspectives (0)
Relationships (2)
Order Details[Order ID] <— List o...
Order Details[Sub-Category] <— ...
Roles (0)
Tables (8)
Avg Profit by Category
List of Orders
Order Details
Order Details Summary
Orders Data
Sales target
Sales Target Summary
100%


Untitled - Power BI Desktop
Search
Sign in
File
Home
Help
Total Amount
Collapse
Order Details
Amount
Category
Order ID
Profit
Profit Margin
Profit Status
Quantity
Collapse
Avg Profit by Category
Average Profit
Category
Collapse
Orders Data
Amount
Category
CustomerName
Location
Order Date
Order ID
Sales target
Category
Month of Order Date
Target
Collapse
All tables
Properties
Relationship
Table
Column
Order Details
Category
Cardinality
Many to one (*:1)
Table
Column
Avg Profit by C...
Category
Make this relationship active
Yes
Cross-filter direction
Single
Apply security filter in both directions
No
Apply changes
Open relationship editor
Data
Tables
Model
Search
Semantic model
Calculation groups (0
Cultures (1)
Expressions (0)
Measures (0)
Perspectives (0)
Relationships (1)
Order Details[Ca
Roles (0)
Tables (8)
Avg Profit by Ca
List of Orders
Order Details
Order Details Su
Orders Data
Sales target
Sales Target Sun
Columns (2)

Data Modeling Logic

Relationships were established between the tables to enable accurate reporting and filtering across datasets. The List of Orders table was connected with the Order Details table using Order ID, while the Order Details table was connected with the Sales Target table using Category.

The relationships were configured and verified using the Manage Relationships feature in Power BI to ensure they were active and correctly defined. Proper cardinality and cross-filter directions were selected based on business logic to maintain data integrity and support efficient analysis across related tables.